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(54) **GRATING STRUCTURES FOR COLOR WAVEGUIDES**

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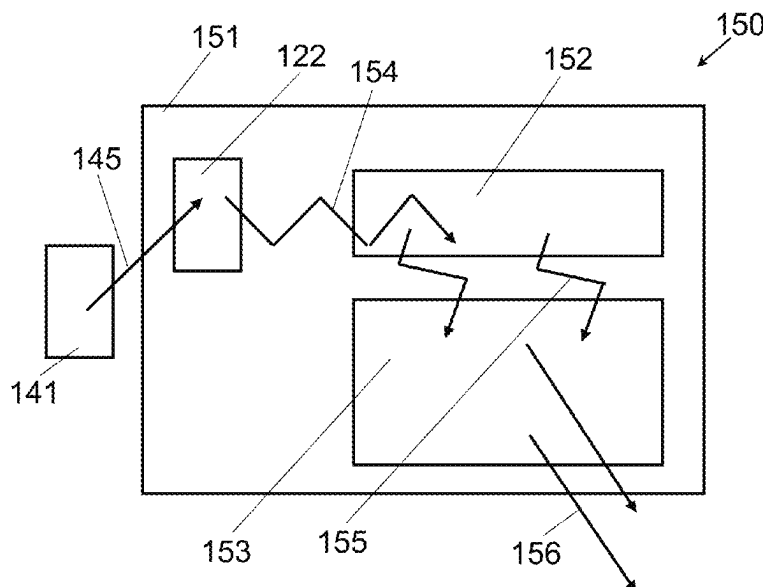
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(57) **ABSTRACT**

Disclosed herein is a waveguide-based display comprising: a waveguide, a source of image modulated light projected over a field of view; an input coupler for coupling said light into a total internal reflection (TIR) path within the waveguide; and a grating structure for providing beam expansion of the TIR light in at least one direction and extracting the TIR light from the waveguide. The grating structure may include a plurality of grating elements having at least two different grating prescriptions and at least one surface relief grating formed by a phase separation process.

19 Claims, 37 Drawing Sheets
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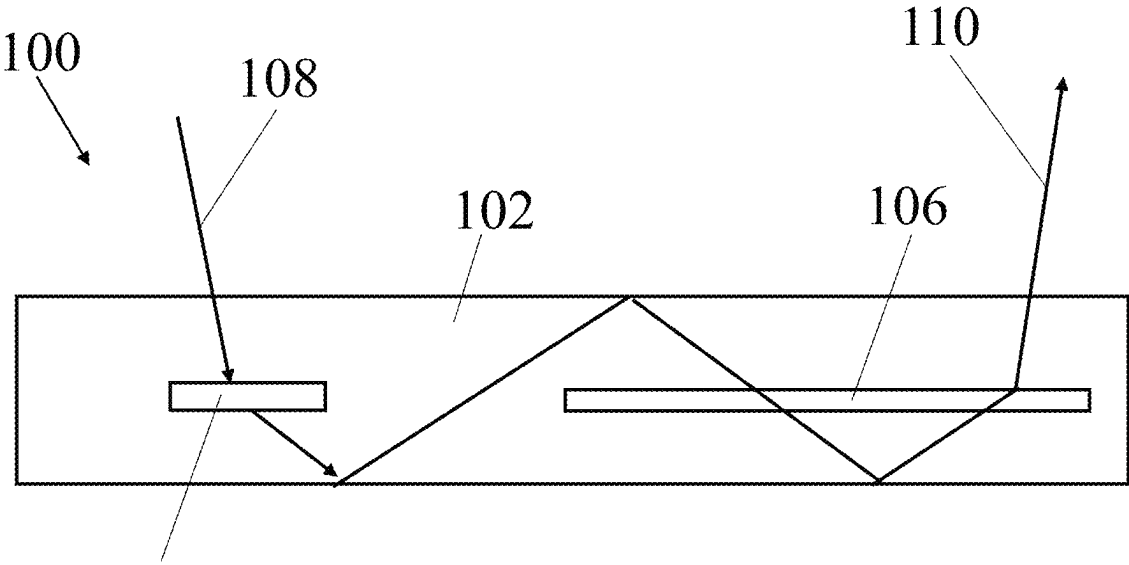


FIG.1A

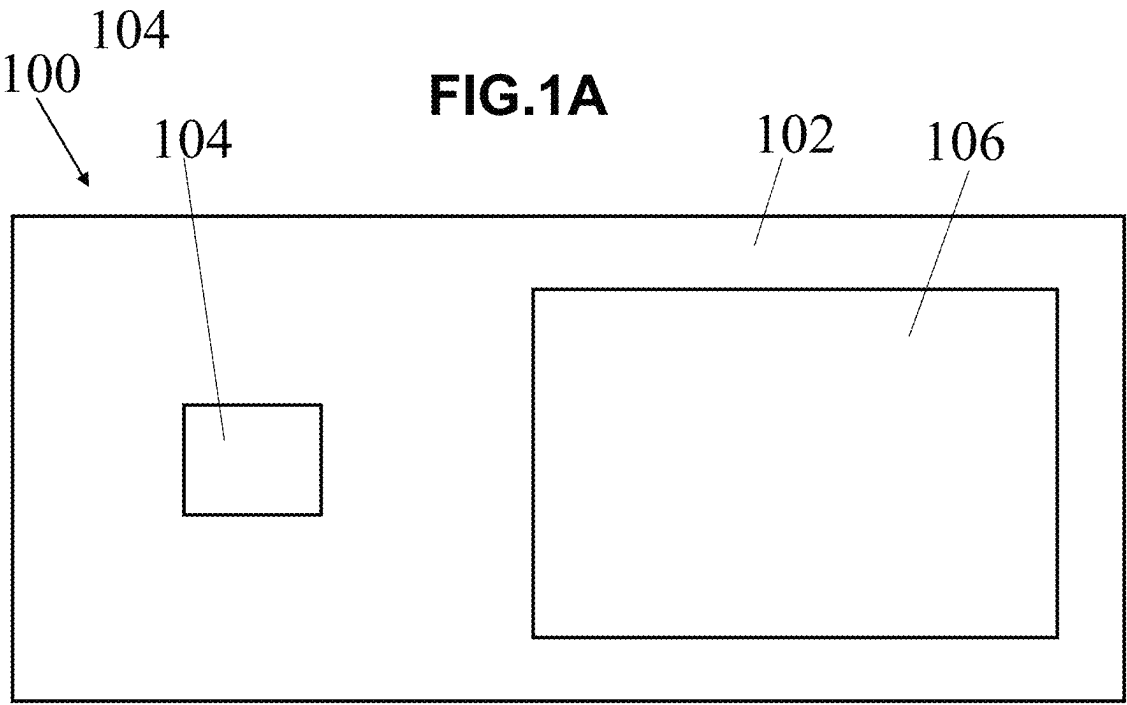
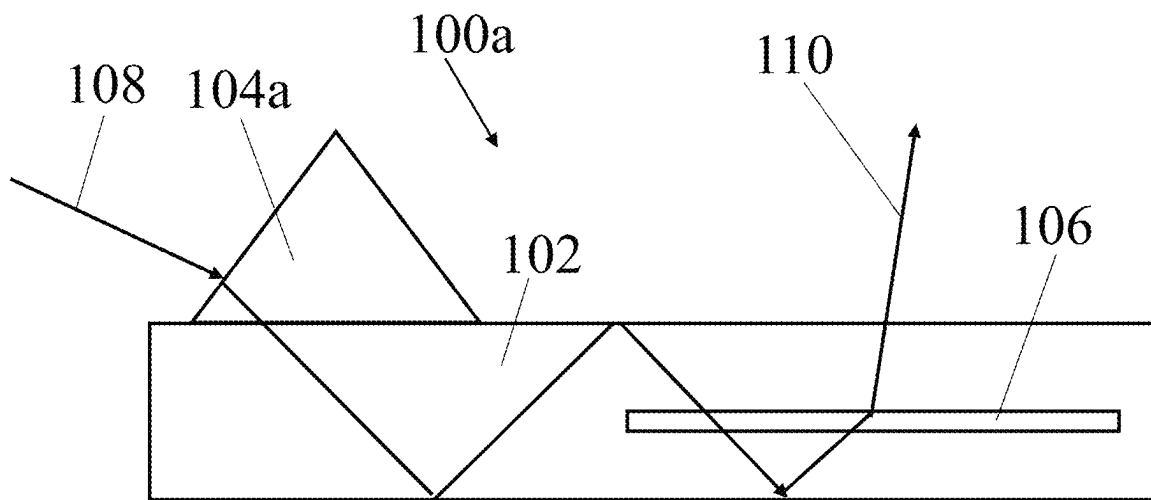


FIG.1B

**FIG.1C**

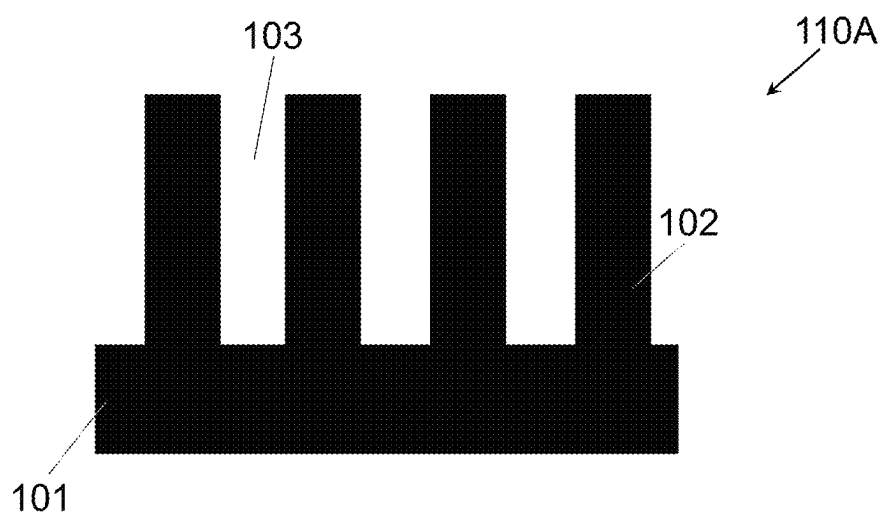


FIG. 2A

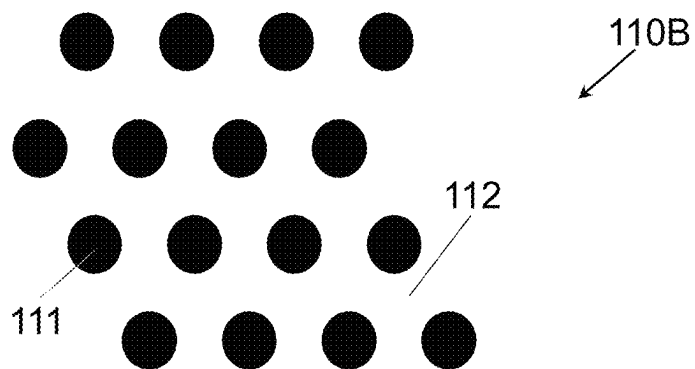


FIG. 2B

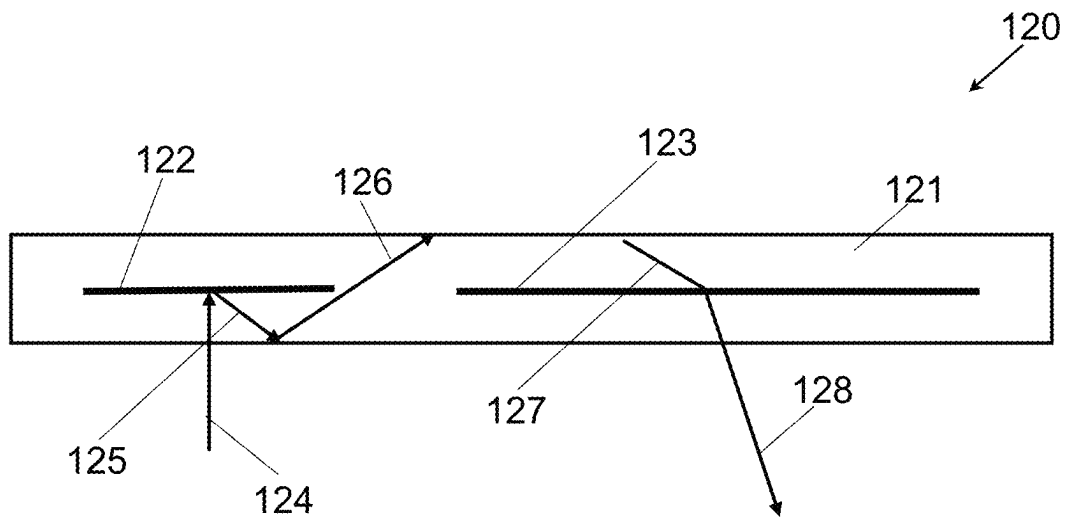


FIG.3

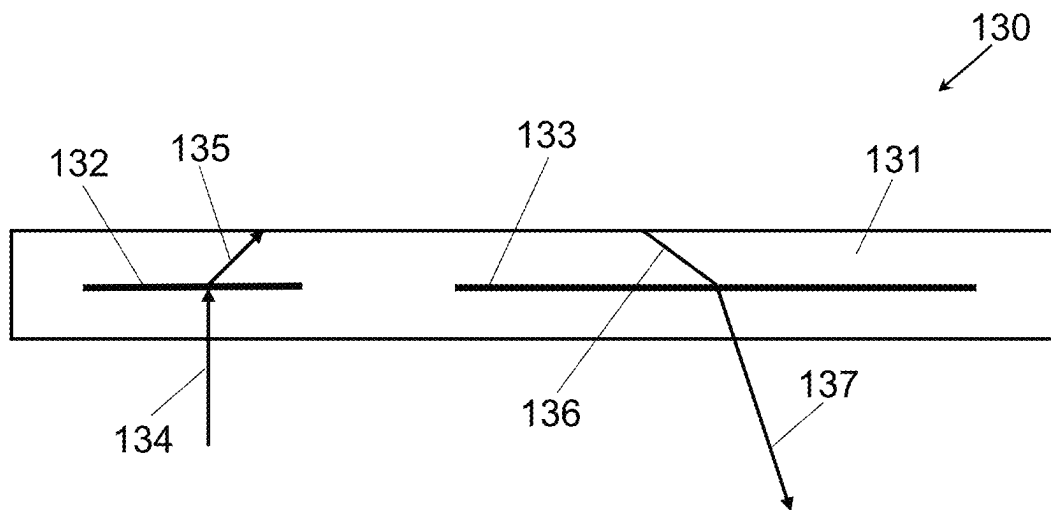


FIG.4

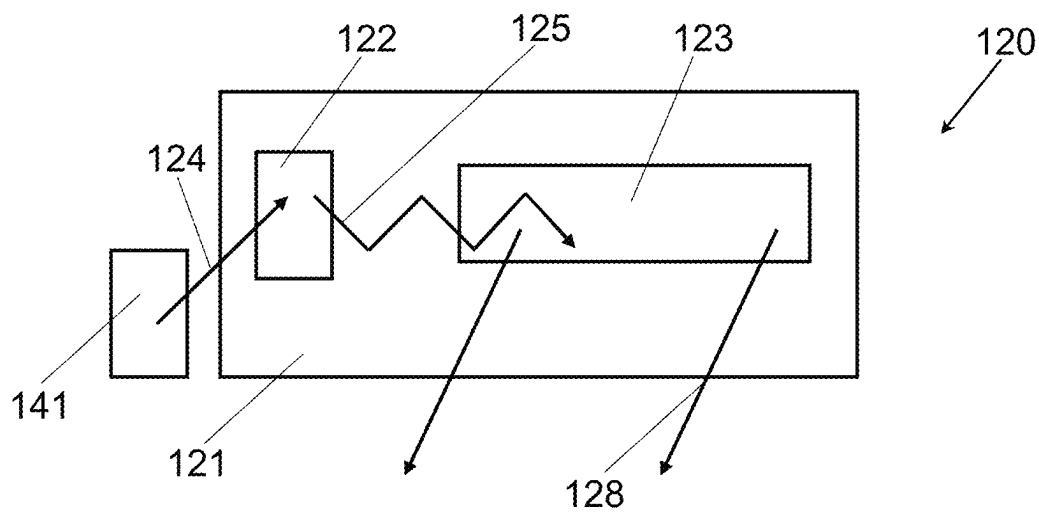


FIG. 5

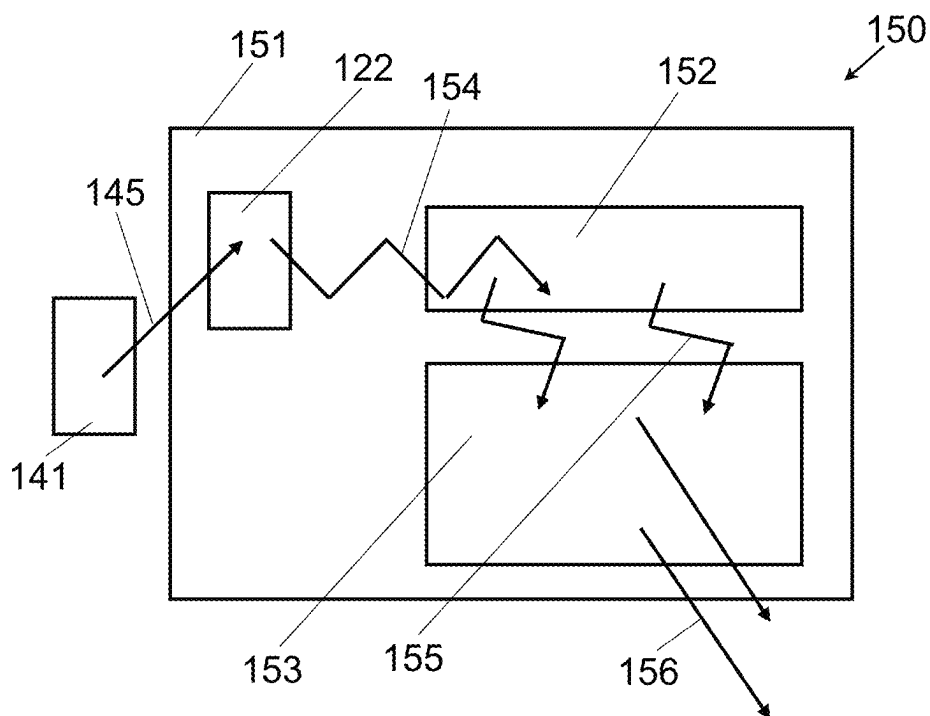
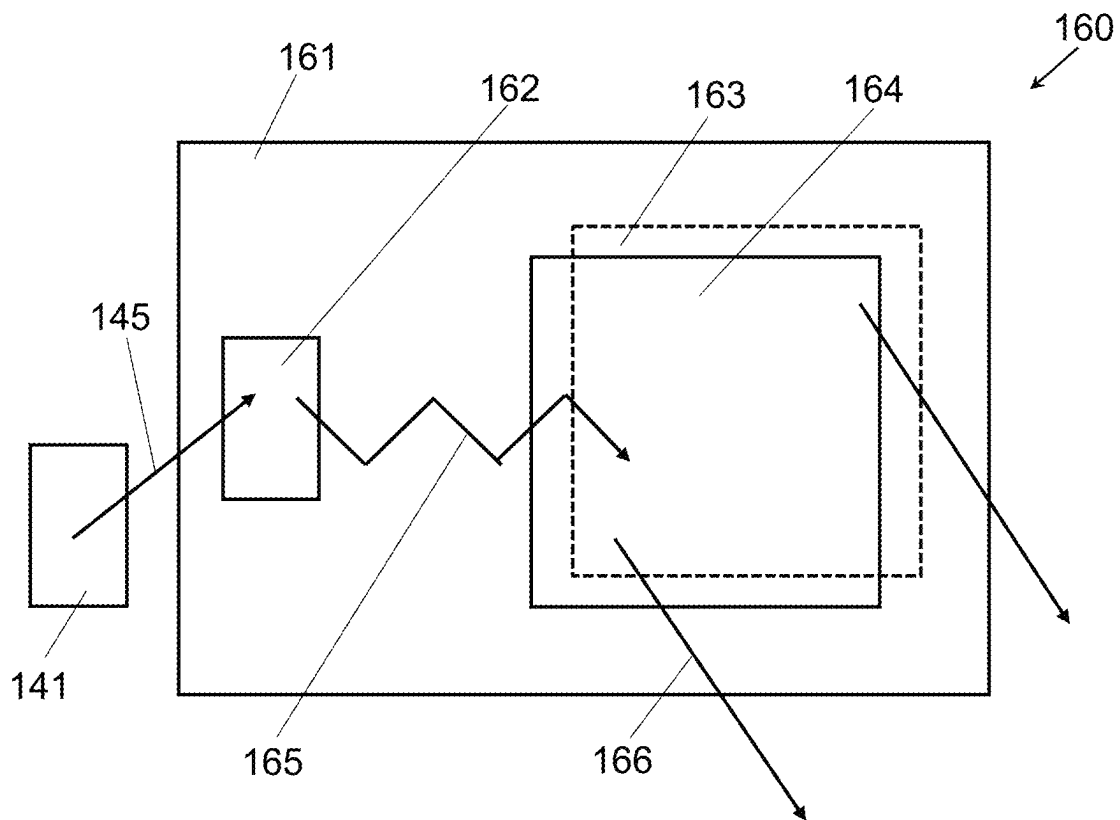


FIG. 6

**FIG. 7**

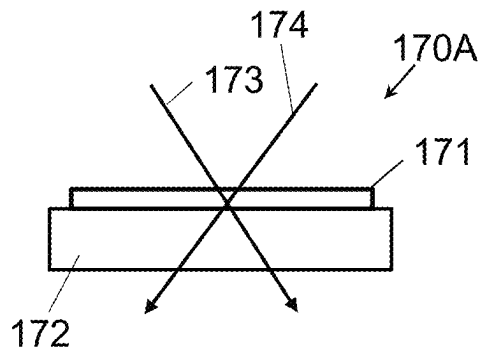


FIG. 8A

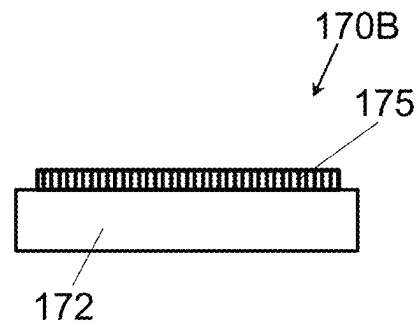


FIG. 8B

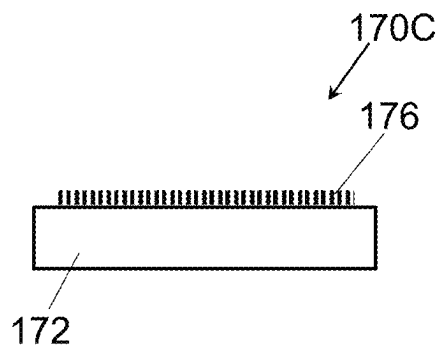


FIG. 8C

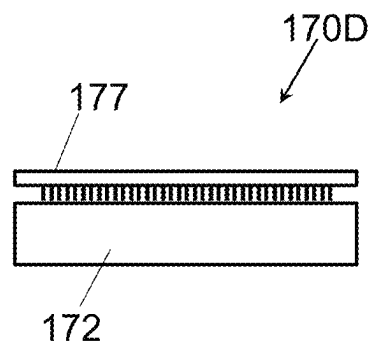


FIG. 8D

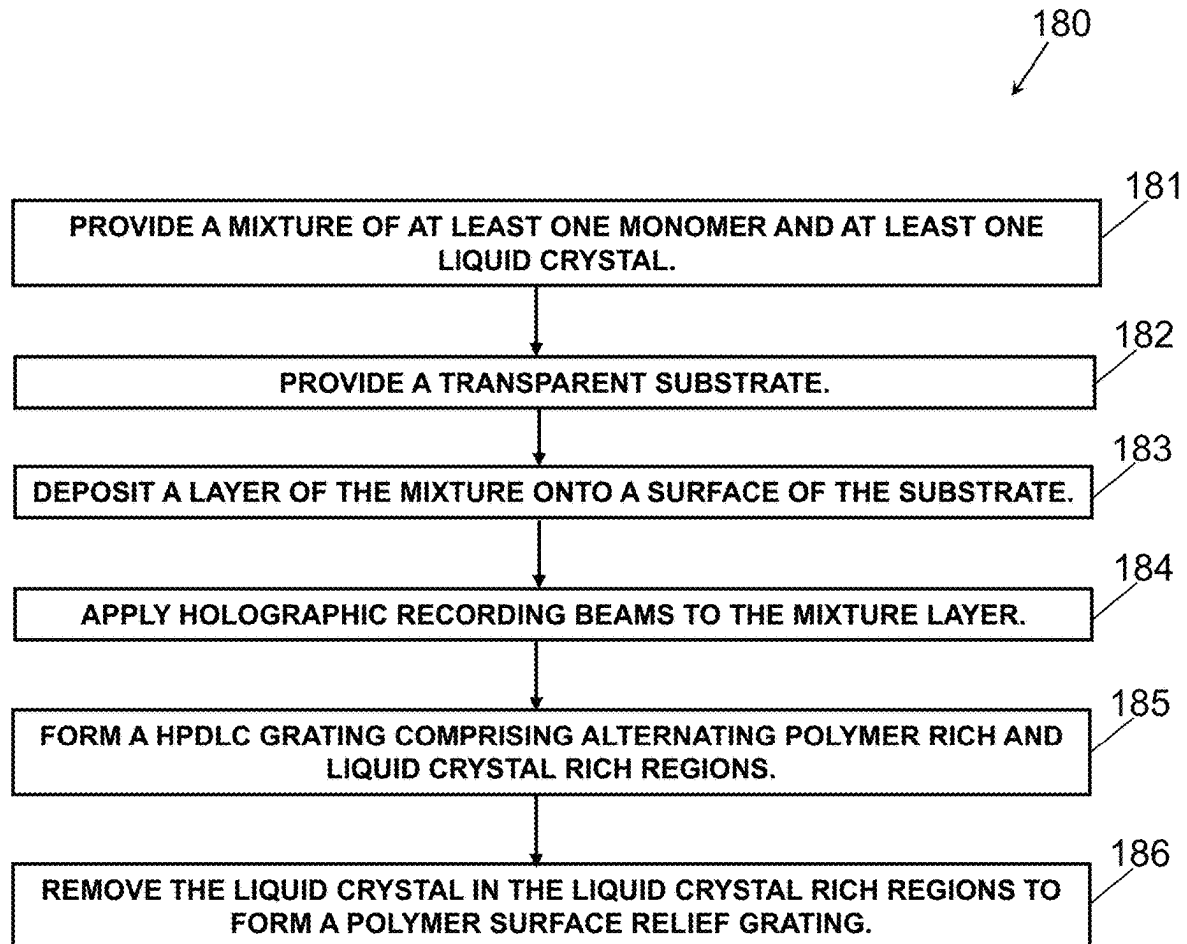


FIG.9

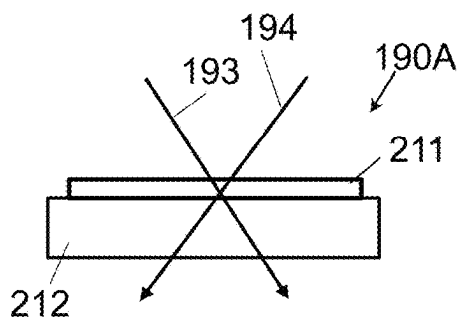


FIG. 10A

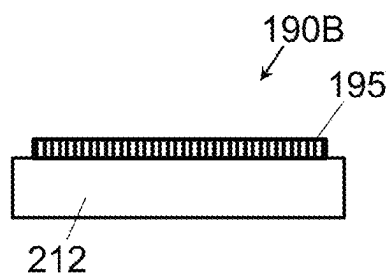


FIG. 10B

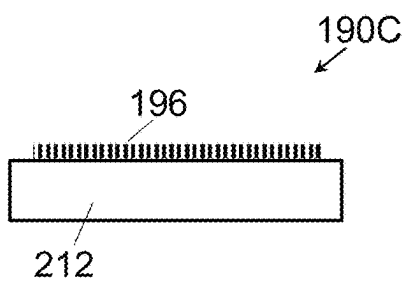


FIG. 10C

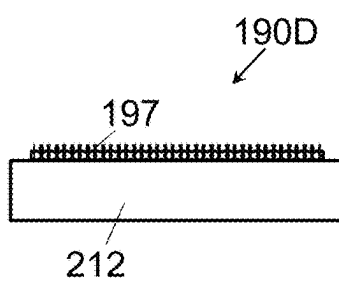


FIG. 10D

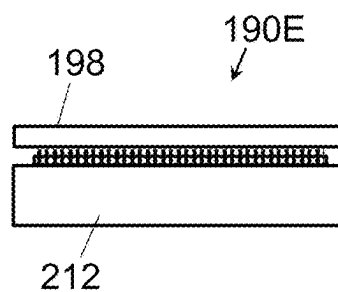


FIG. 10E

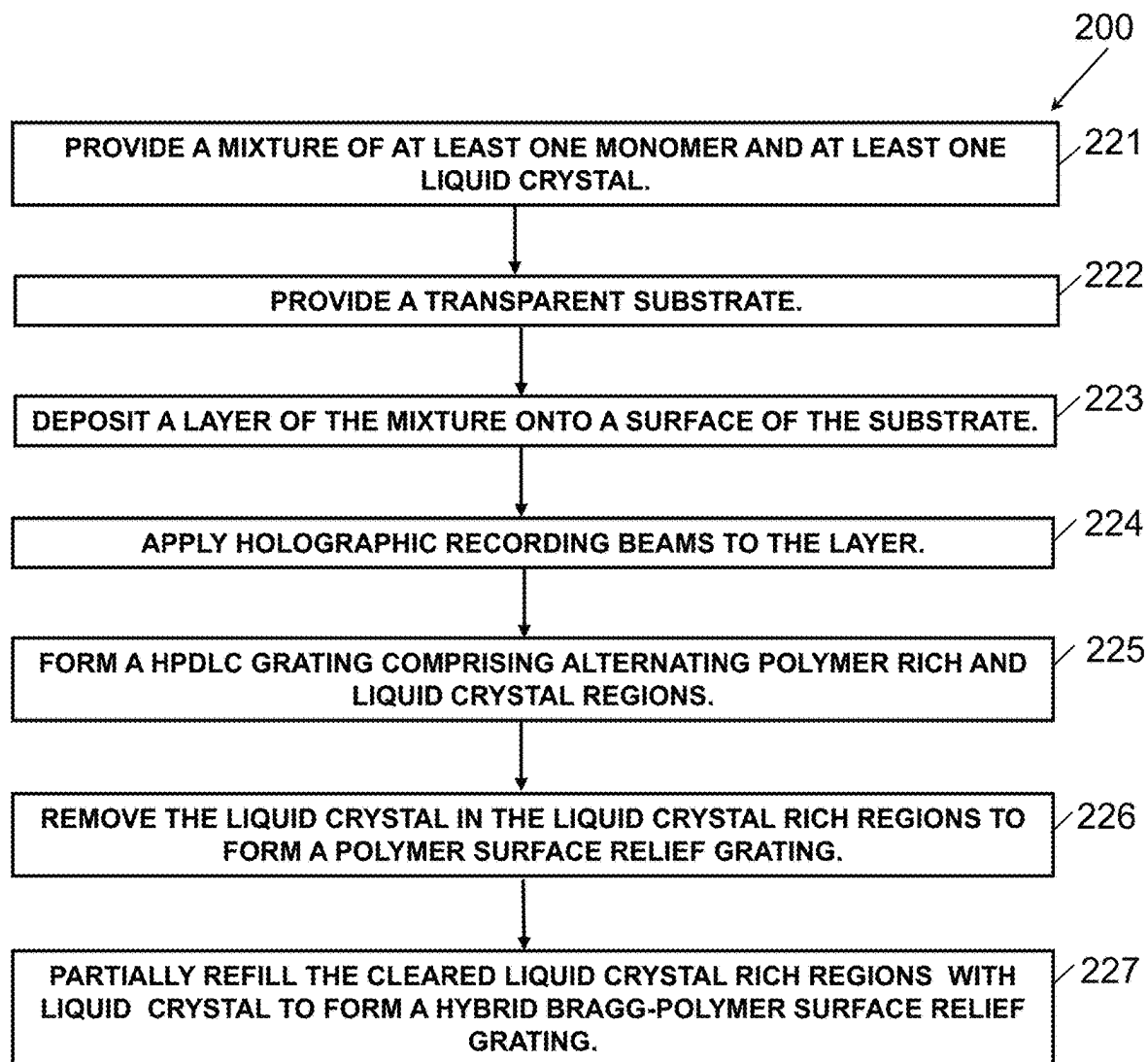


FIG.11

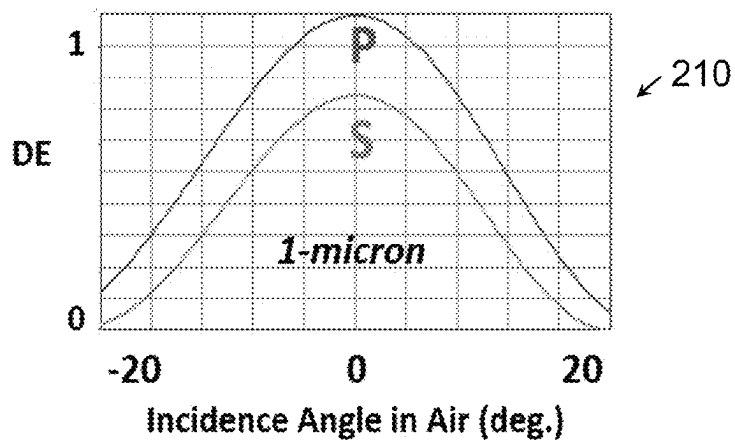


FIG.12

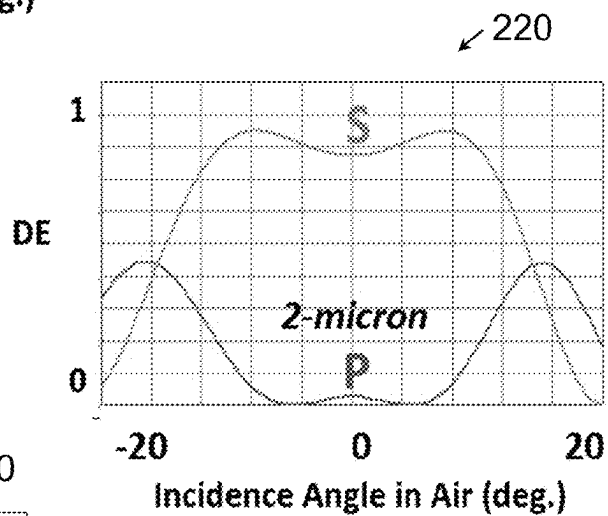


FIG.13

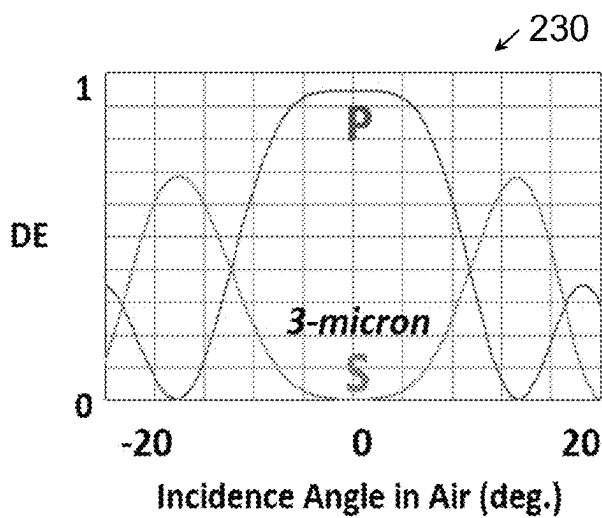
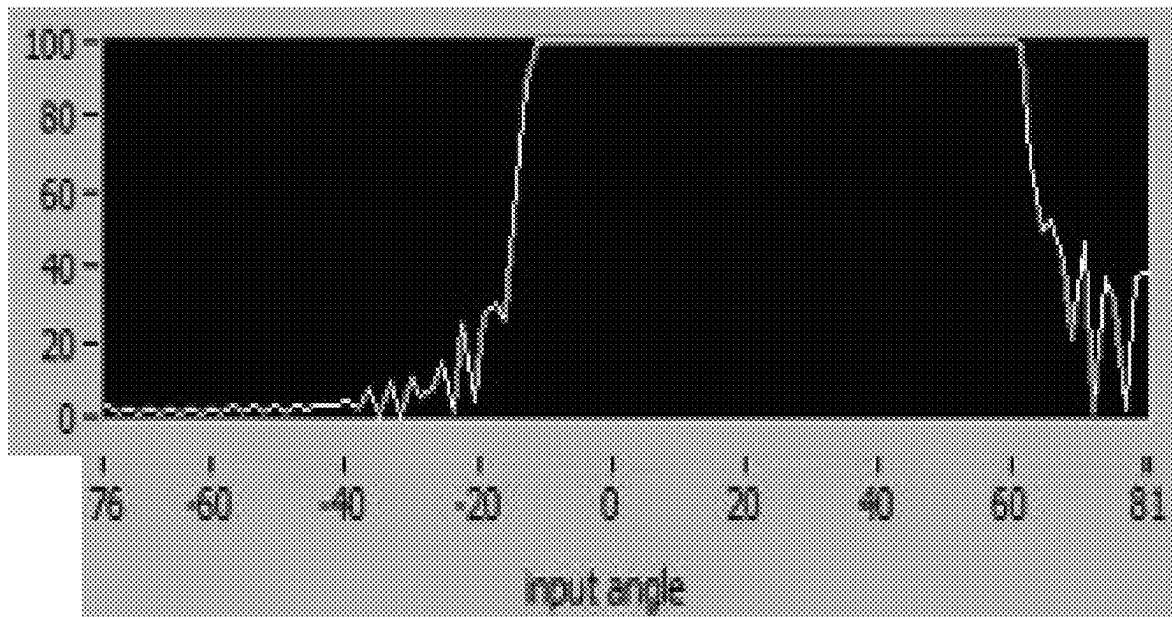
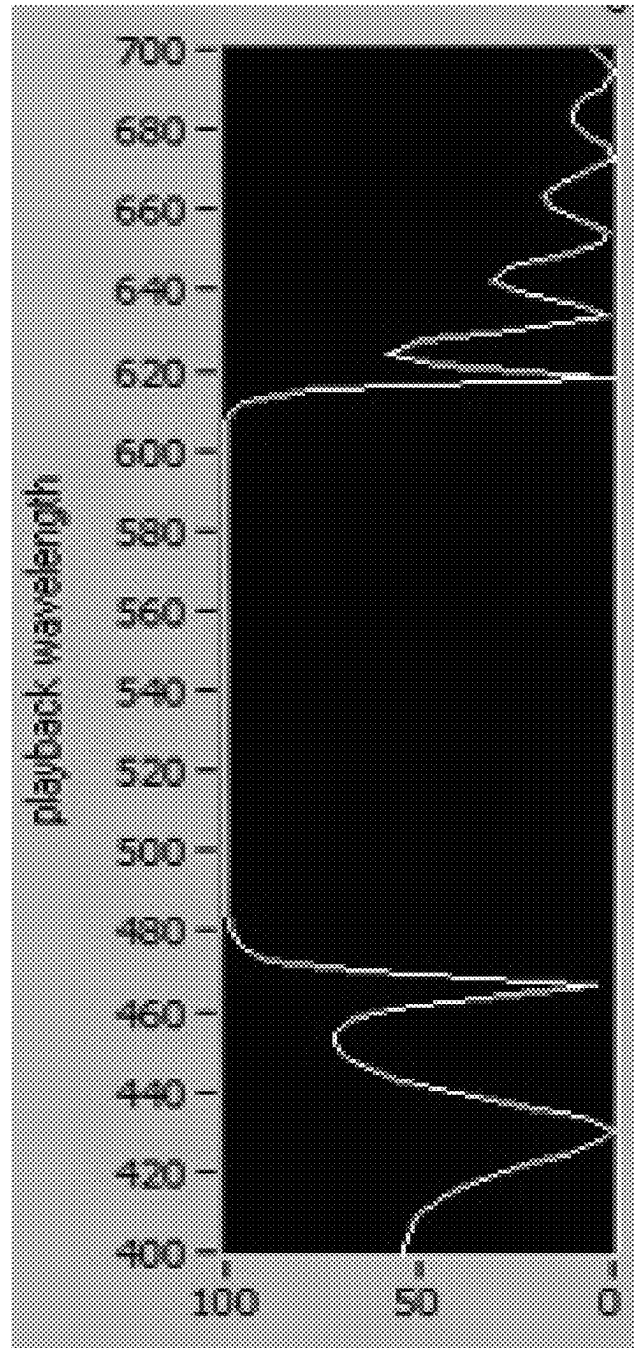
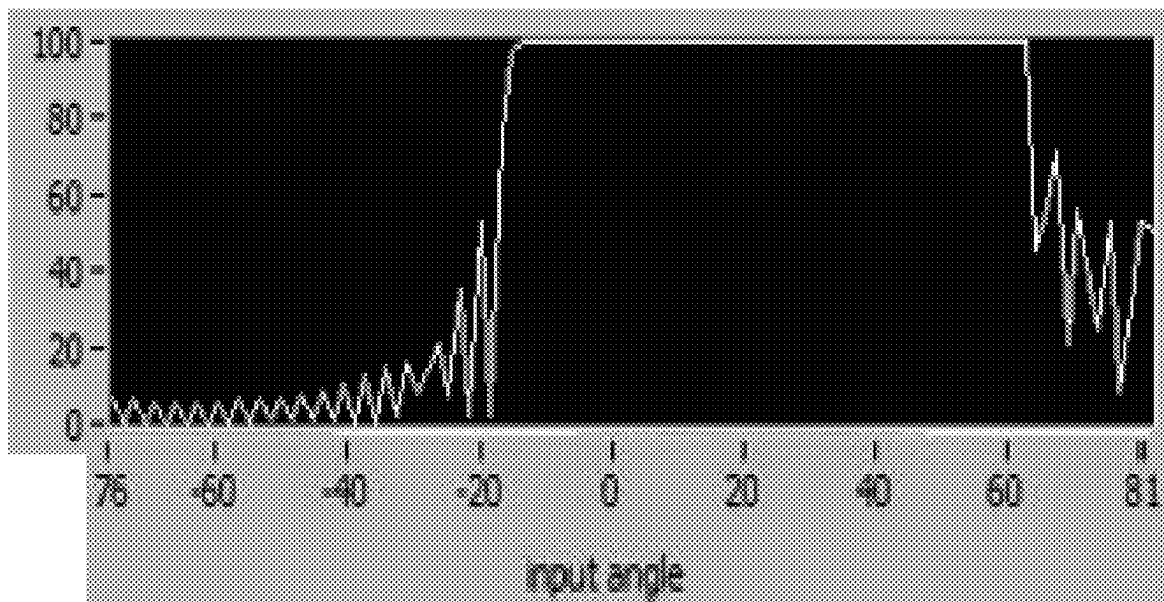


FIG.14

**FIG.15A**

**FIG.15B**

**FIG.16A**

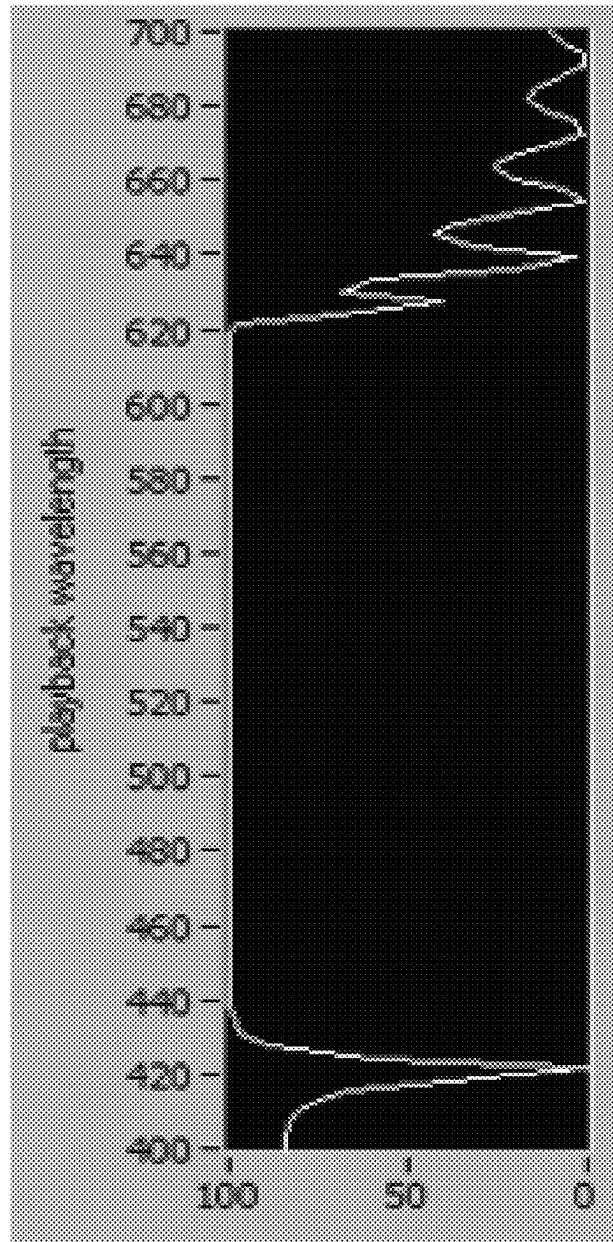


FIG.16B

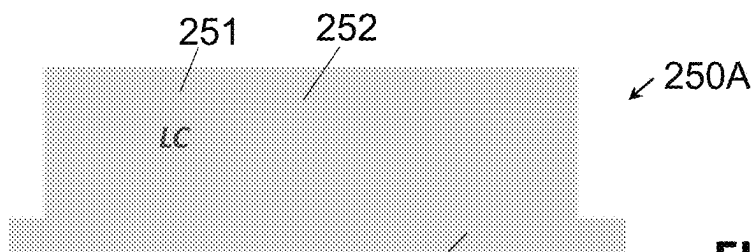


FIG. 17A

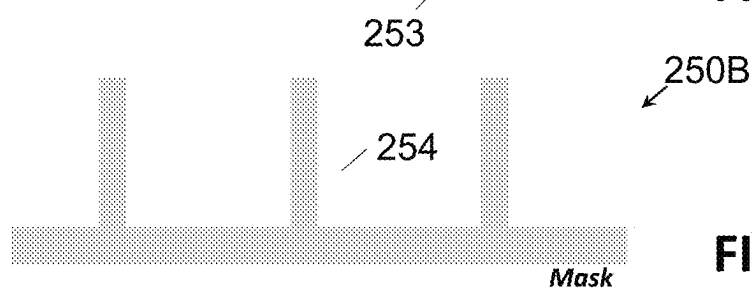


FIG. 17B

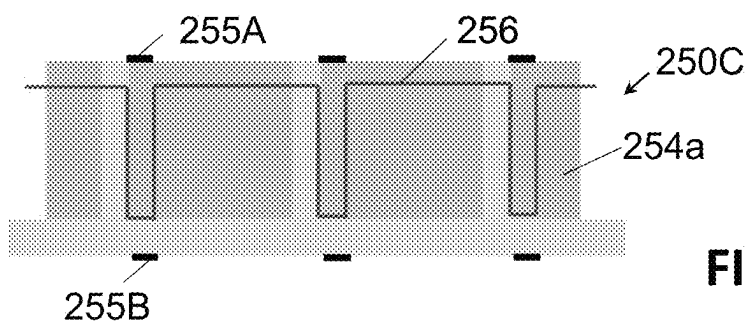


FIG. 17C

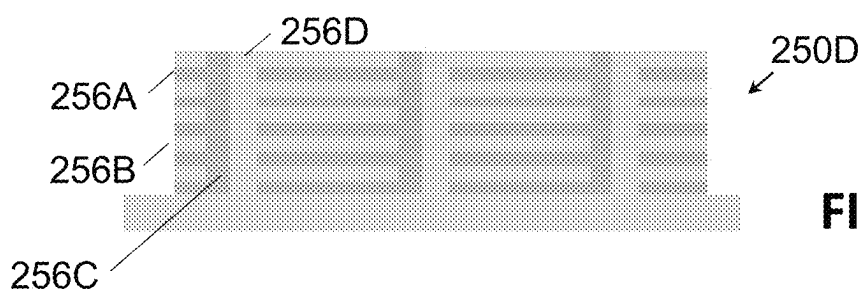


FIG. 17D

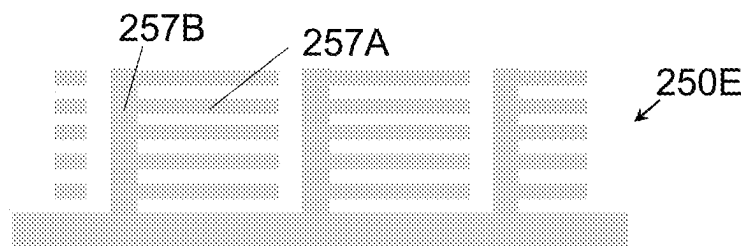


FIG. 17E

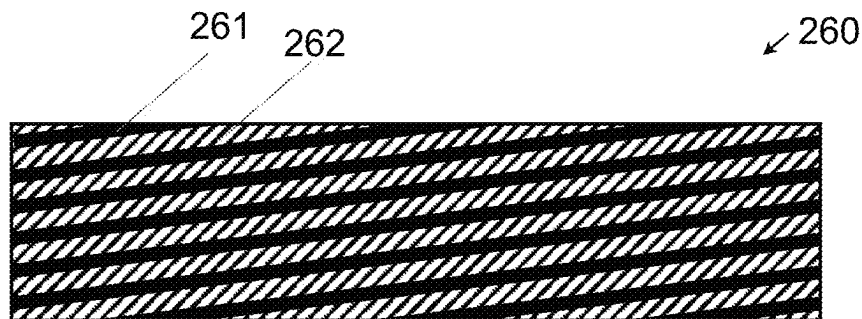


FIG.18

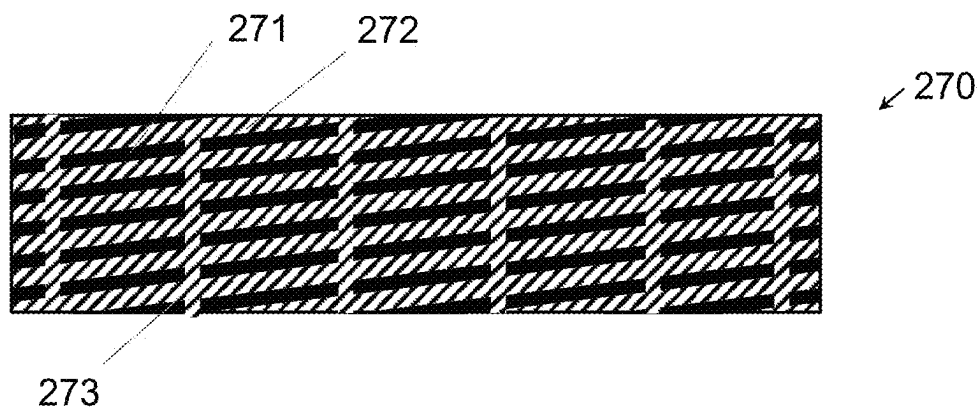


FIG.19

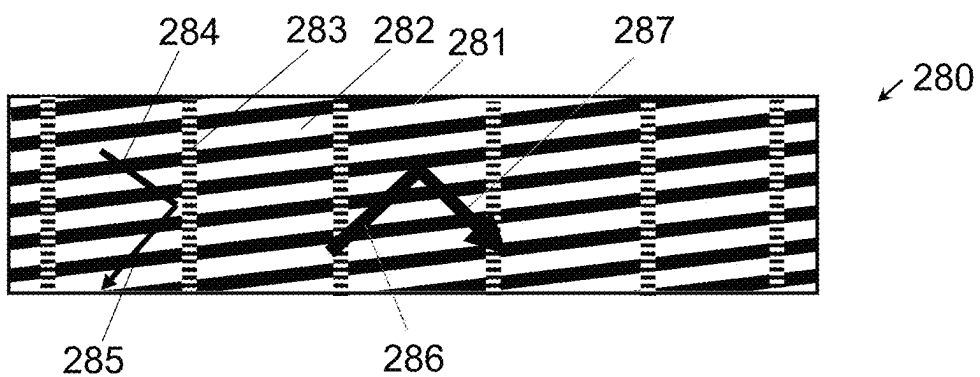
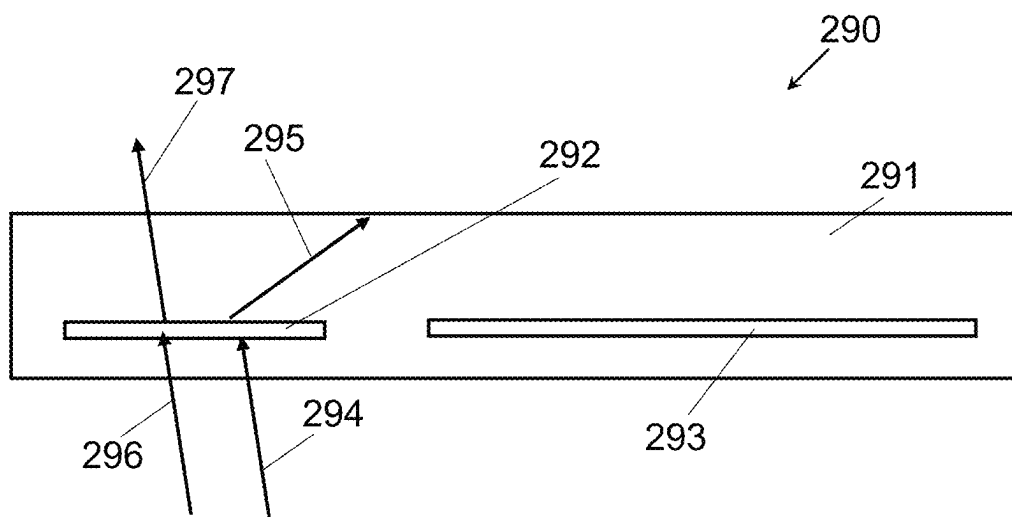
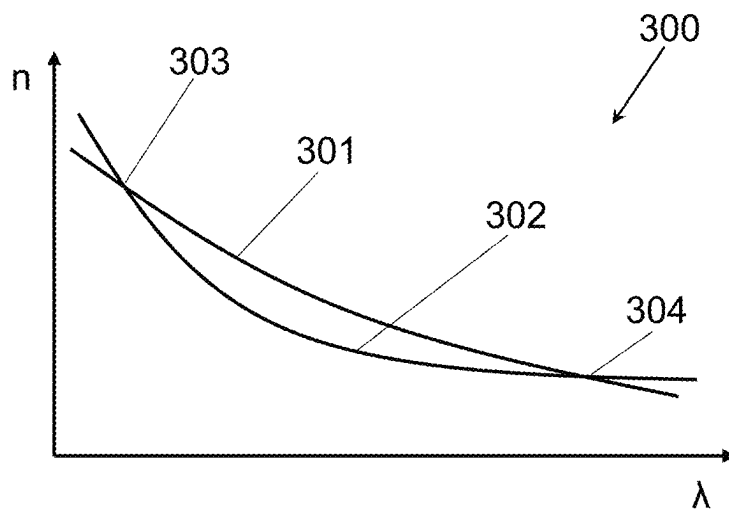


FIG.20

**FIG.21****FIG.22**

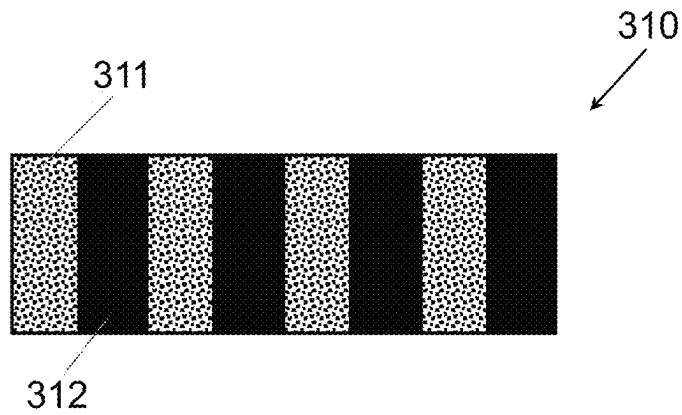


FIG.23

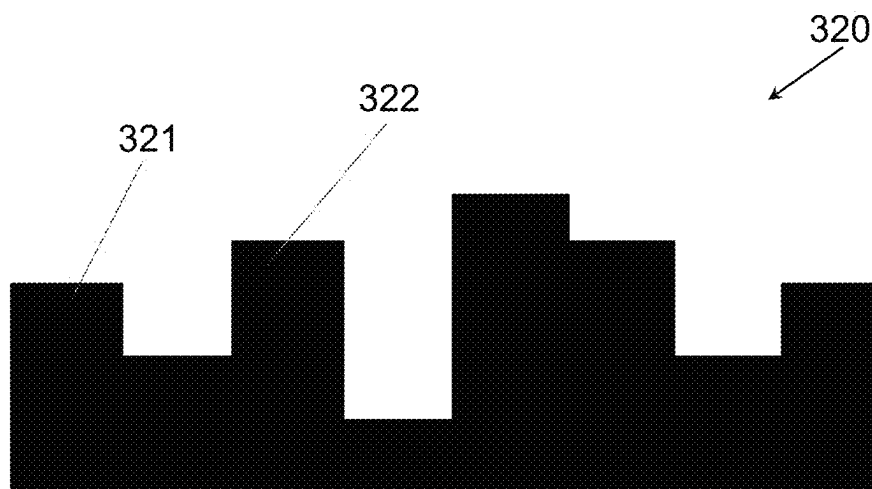


FIG.24

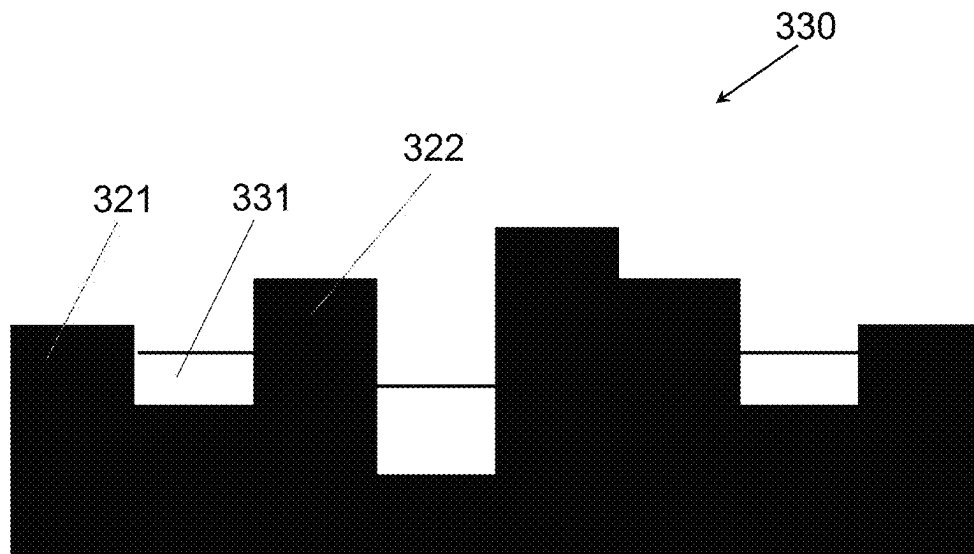


FIG.25

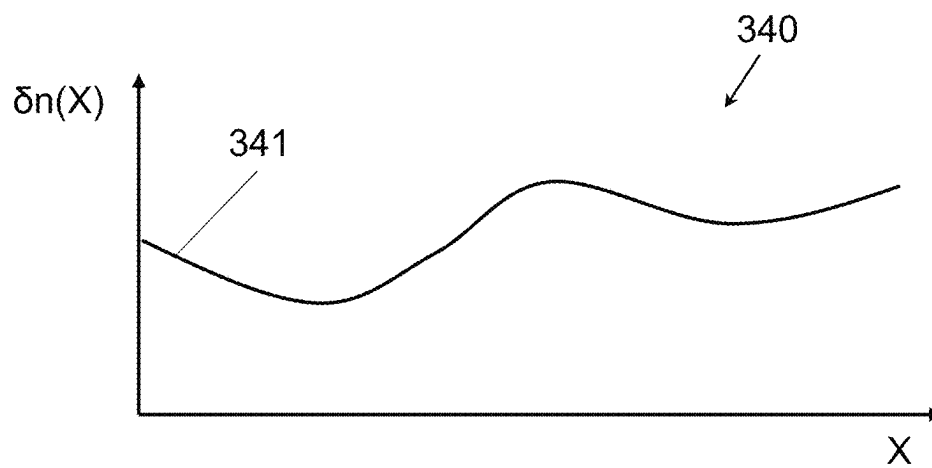
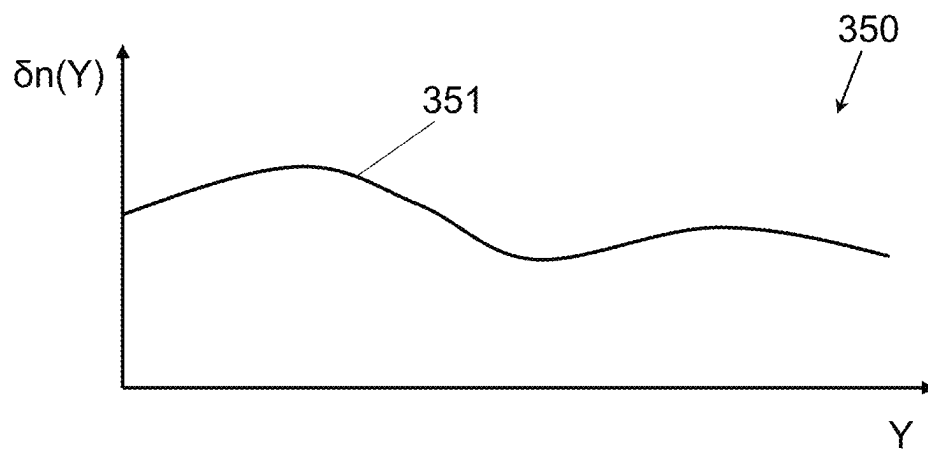


FIG.26

**FIG.27**

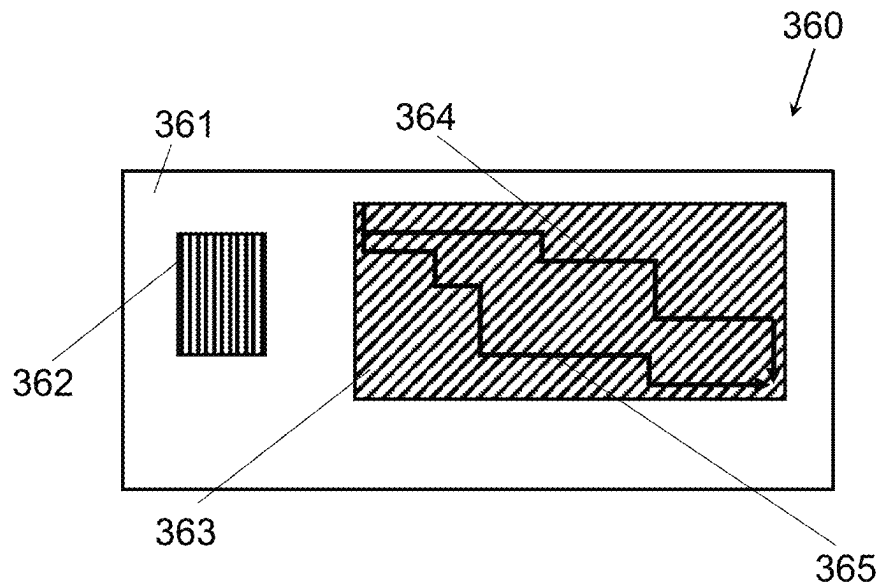


FIG.28

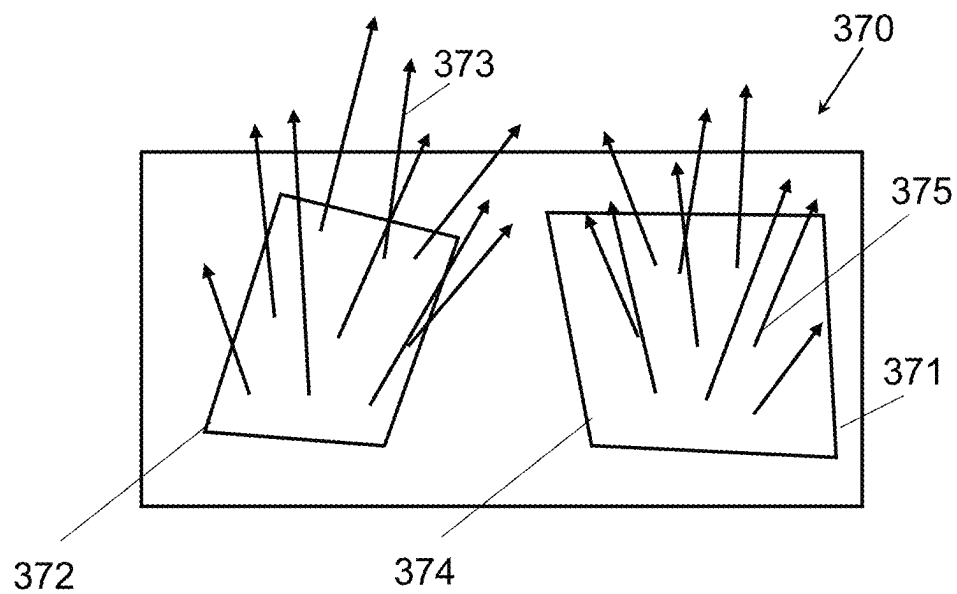
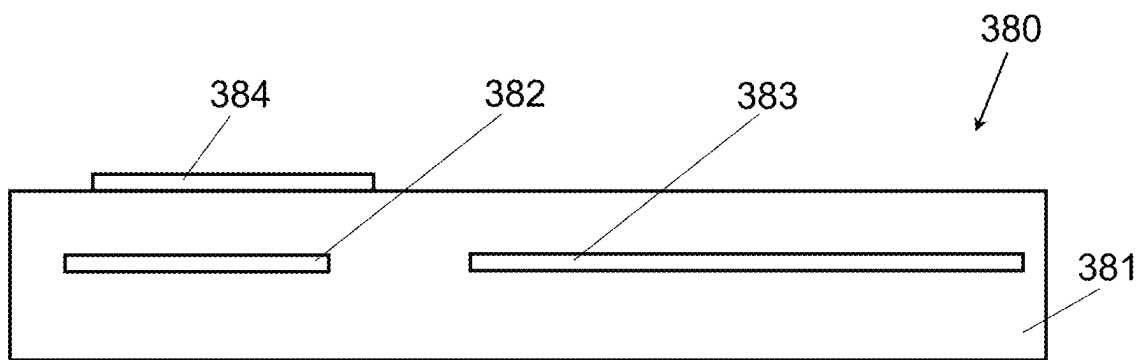


FIG.29

**FIG.30**

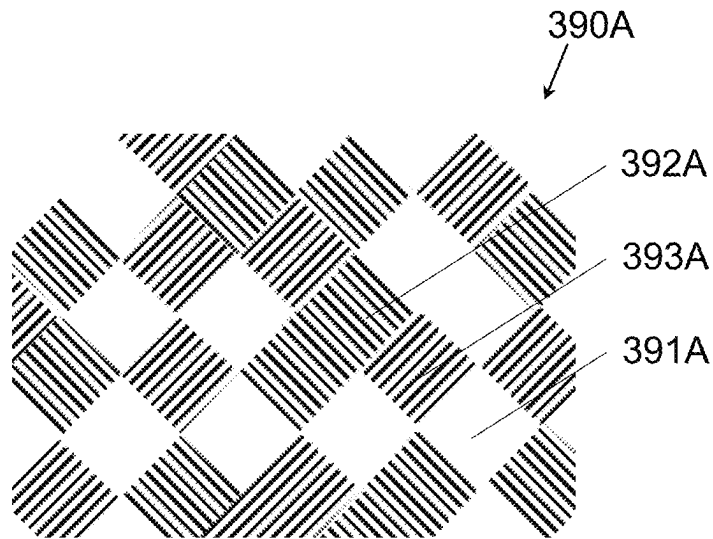


FIG. 31

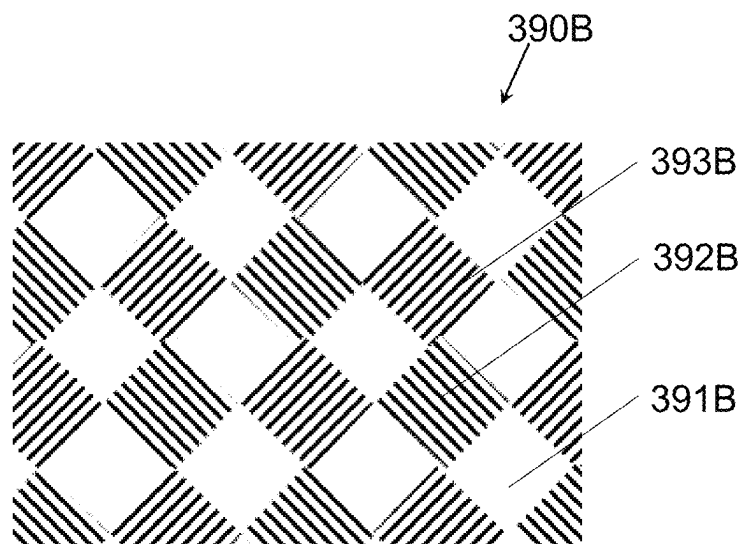
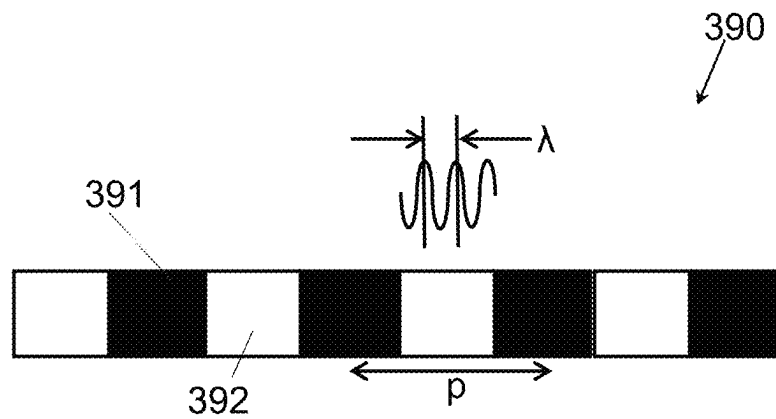


FIG. 32

**FIG.33**

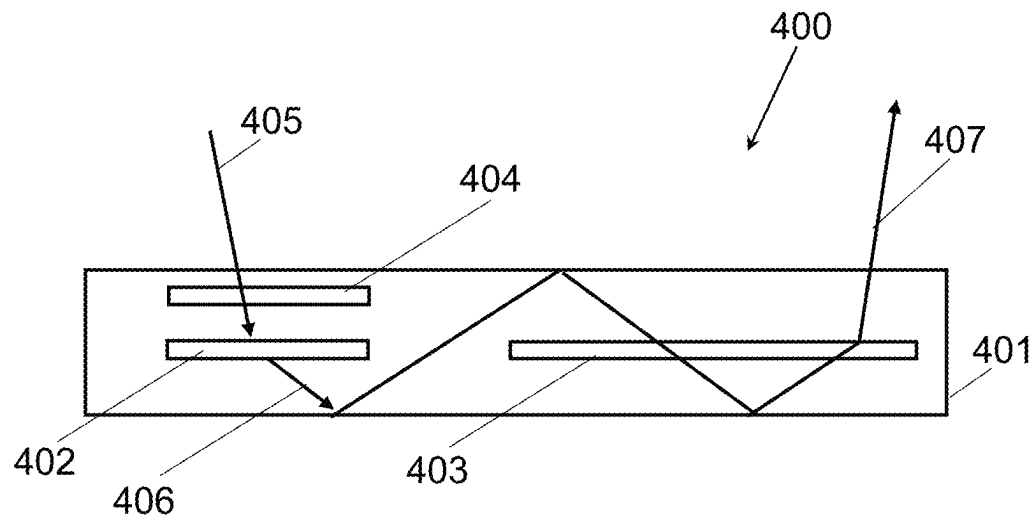


FIG.34

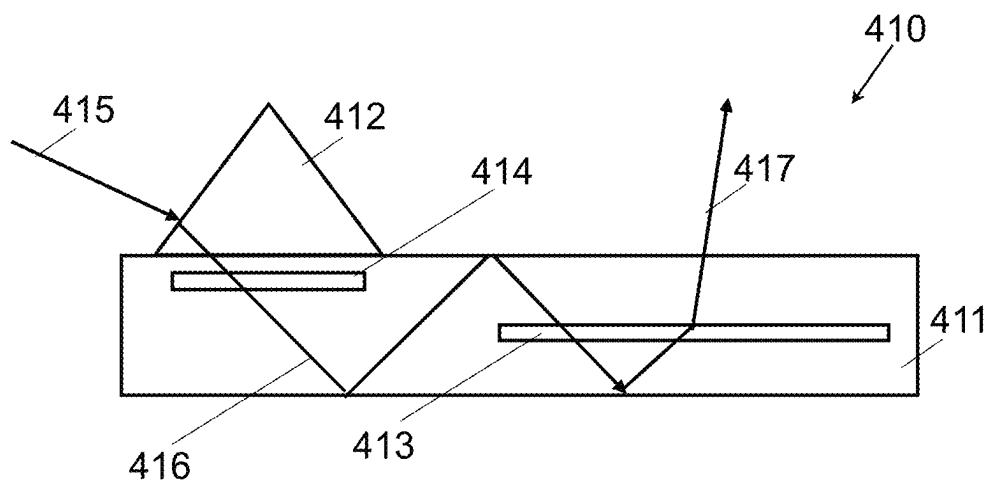


FIG.35

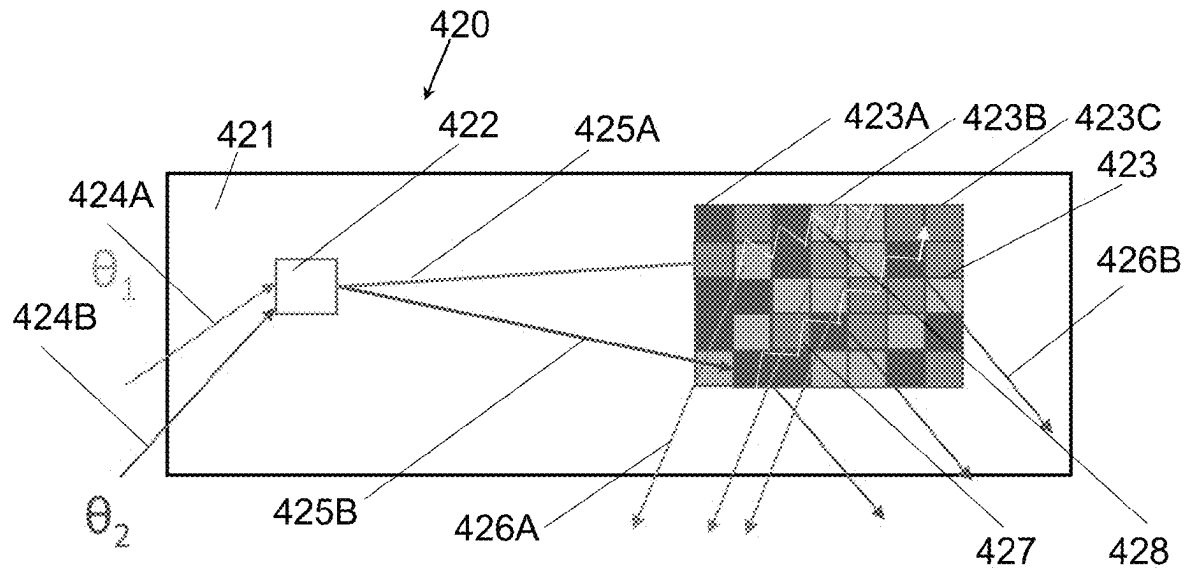


FIG.36A

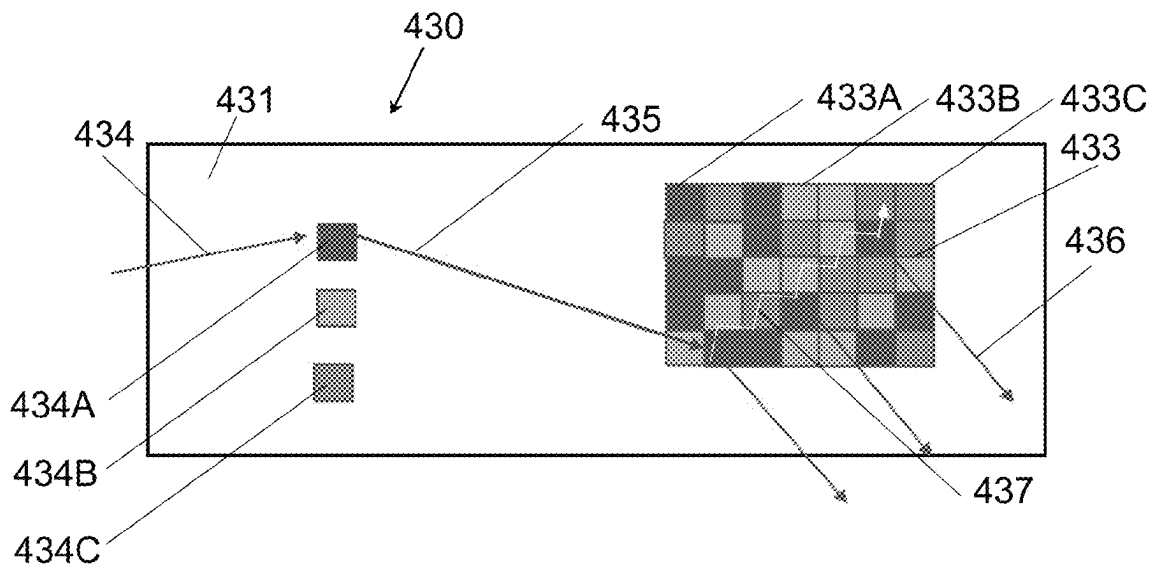
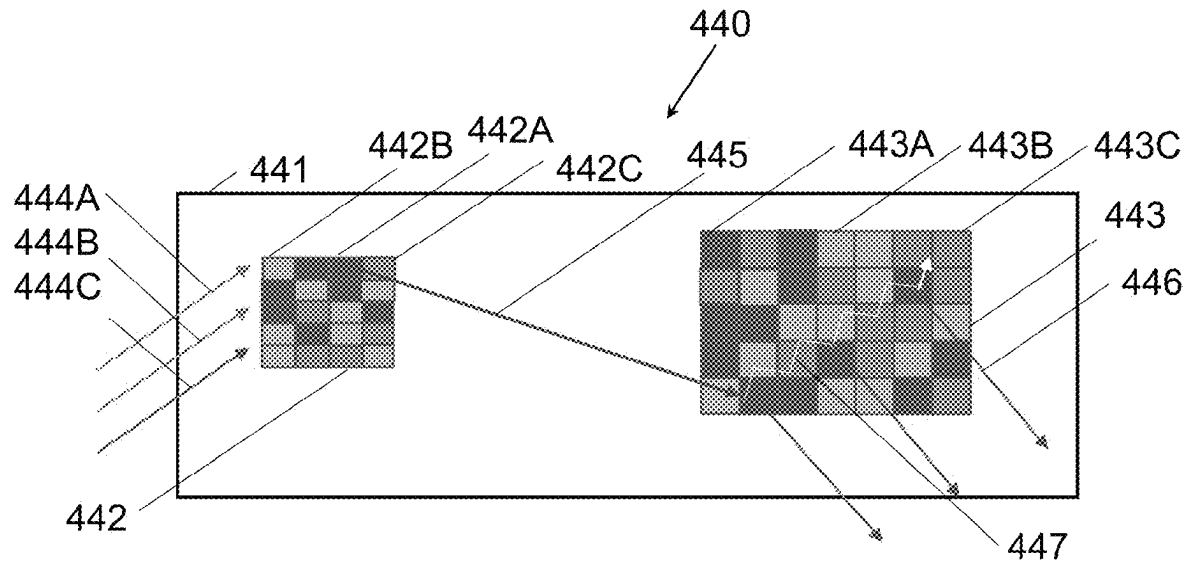
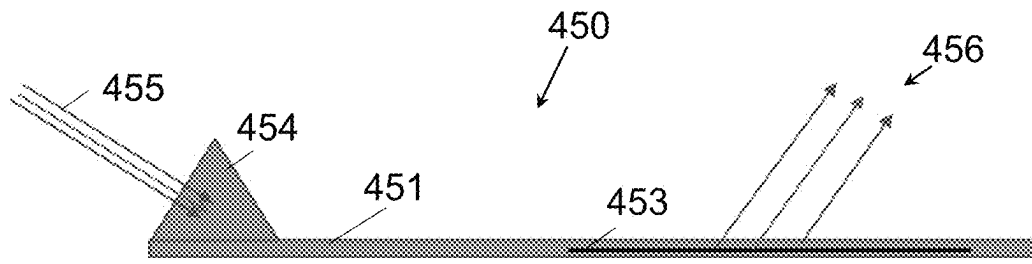
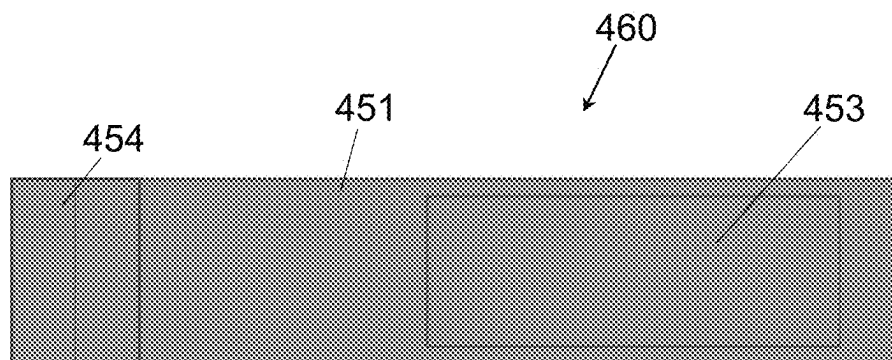


FIG.36B

**FIG.36C**

**FIG.37****FIG.38**

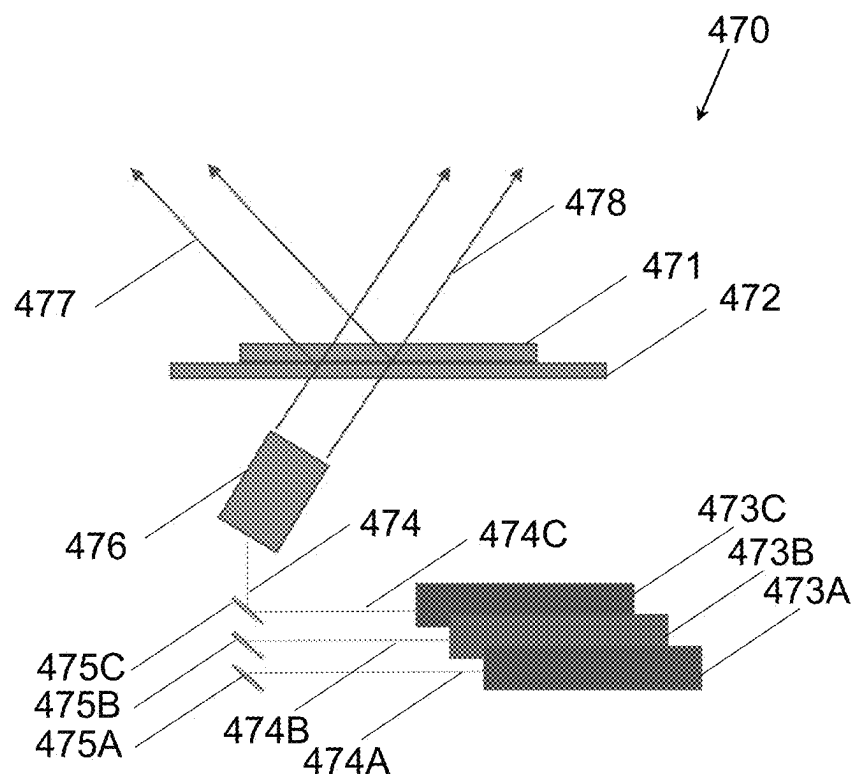


FIG.39

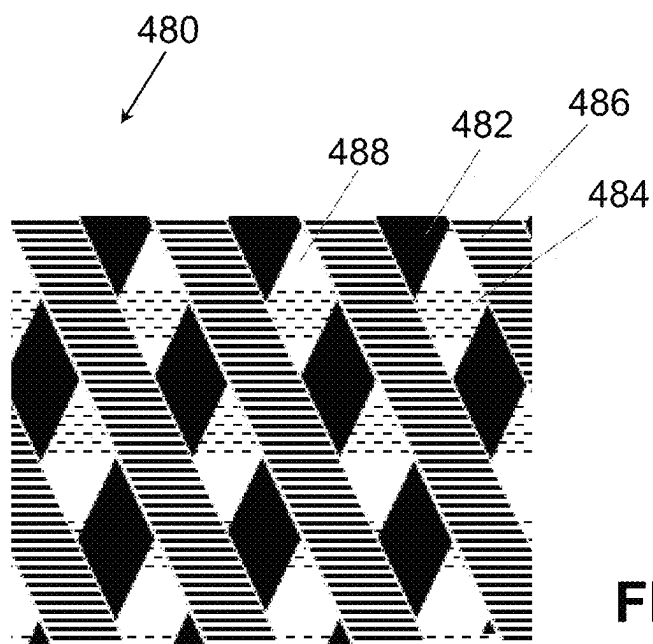


FIG. 40

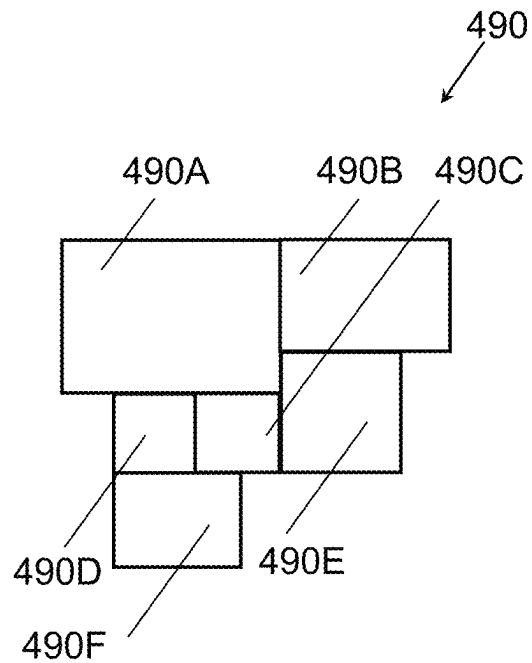


FIG. 41A

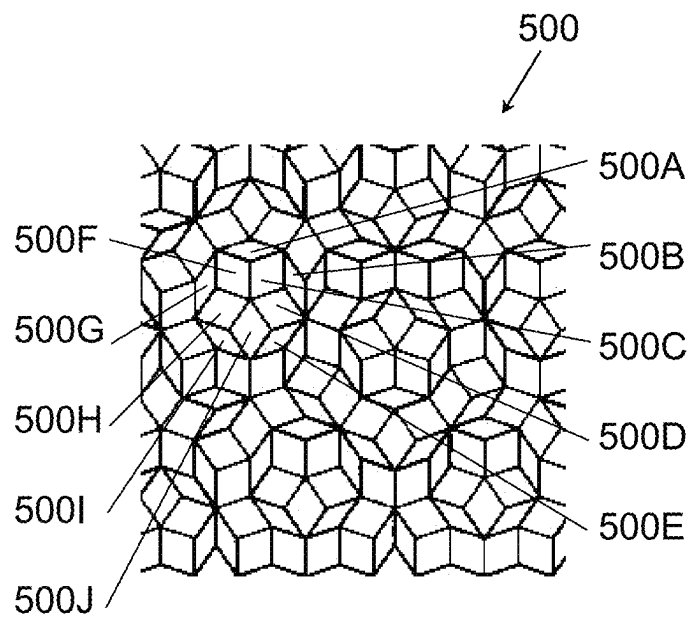


FIG. 41B

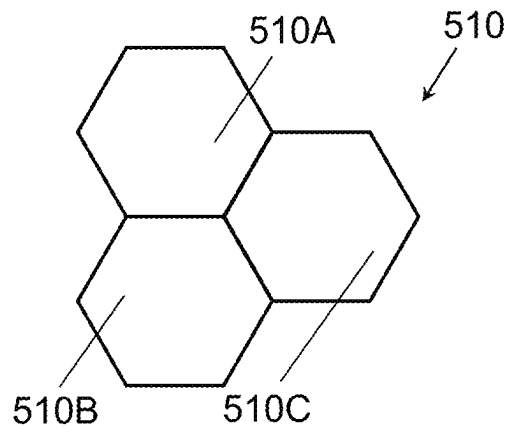


FIG.41C

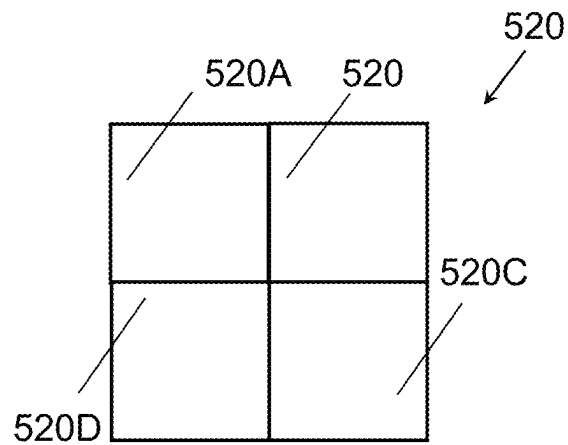


FIG.41D

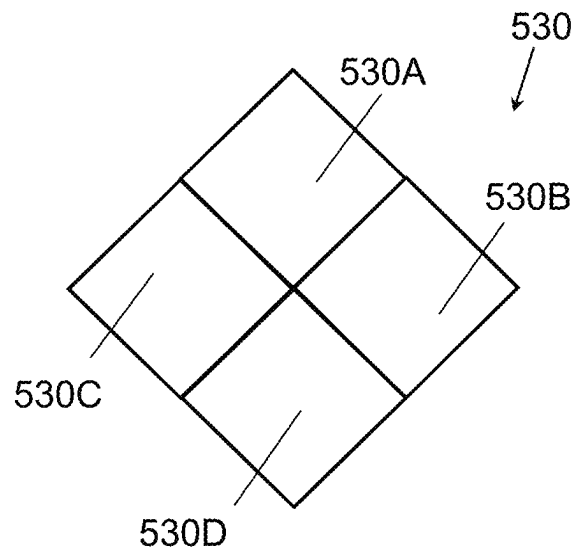


FIG.41E

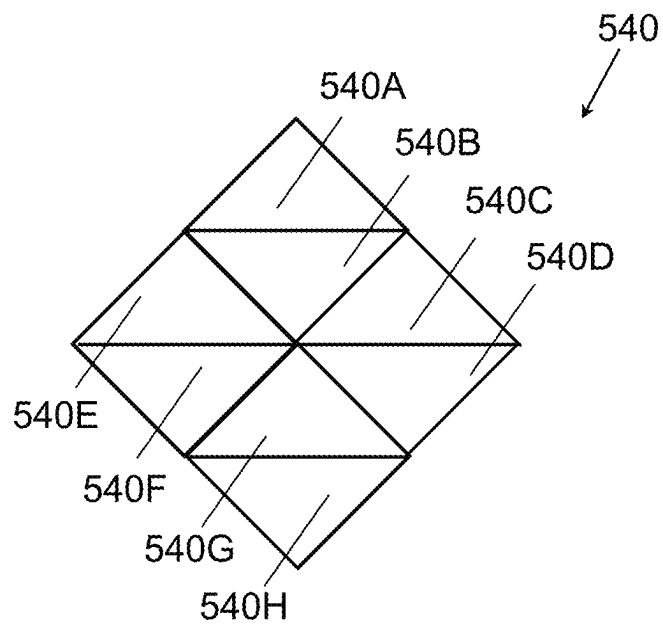


FIG.41F

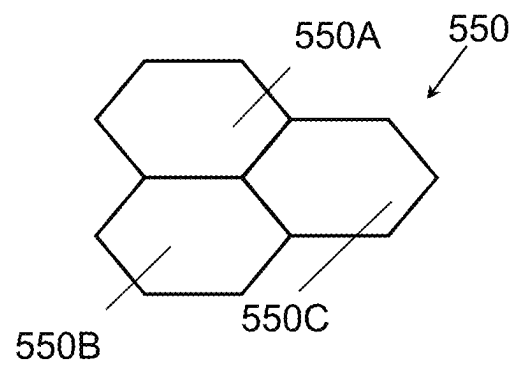


FIG.41G

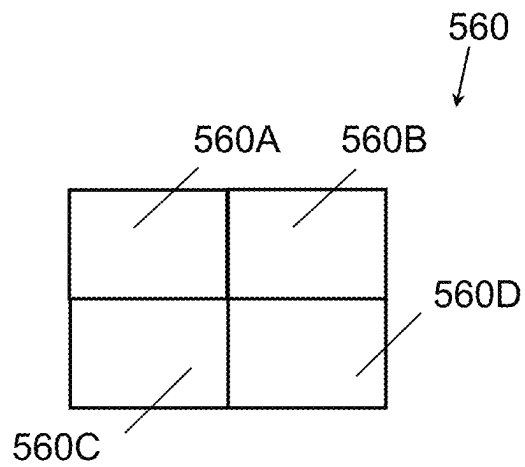


FIG.41H

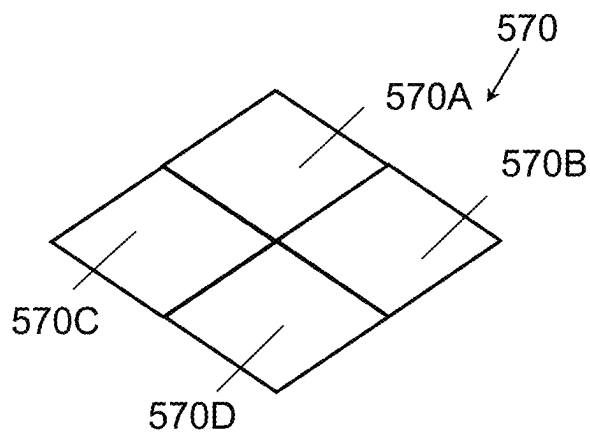


FIG.41I

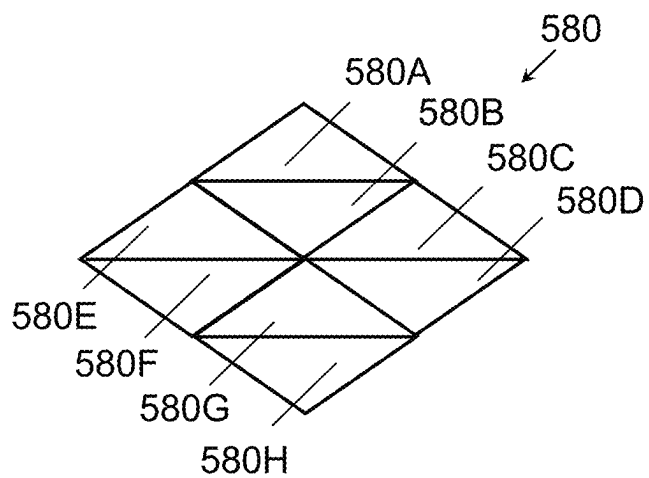
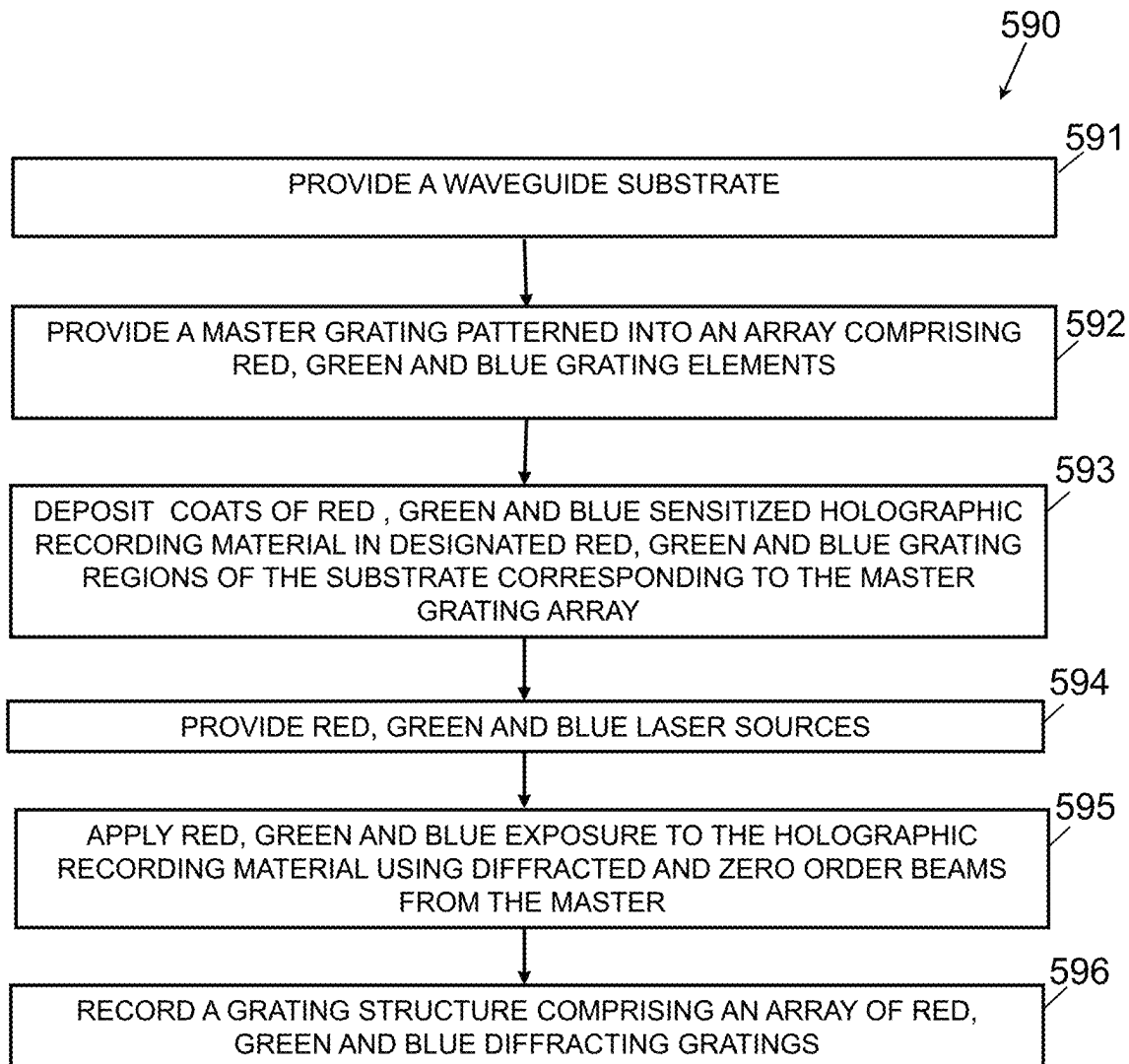


FIG.41J

**FIG.42**

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GRATING STRUCTURES FOR COLOR WAVEGUIDES

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of and priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application Ser. No. 63/134,898, entitled “Grating Structures for Color Waveguides” and filed Jan. 7, 2021, which is incorporated herein by reference in its entirety for all purposes.

FIELD OF THE INVENTION

The present invention generally relates to grating structures for providing beam expansion and extracting light from a waveguide.

BACKGROUND

Waveguides can be referred to as structures with the capability of confining and guiding waves (e.g., restricting the spatial region in which waves can propagate). One subclass includes optical waveguides, which are structures that can guide electromagnetic waves, typically those in the visible spectrum. Waveguide structures can be designed to control the propagation path of waves using a number of different mechanisms. For example, planar waveguides can be designed to utilize diffraction gratings to diffract and couple incident light into the waveguide structure such that the incoupled light can proceed to travel within the planar structure via total internal reflection (“TIR”).

Fabrication of waveguides can include the use of material systems that allow for the recording of holographic optical elements within the waveguides. One class of such material includes polymer dispersed liquid crystal (“PDLC”) mixtures, which are mixtures containing photopolymerizable monomers and liquid crystals. A further subclass of such mixtures includes holographic polymer dispersed liquid crystal (“HPDLC”) mixtures. Holographic optical elements, such as volume phase gratings, can be recorded in such a liquid mixture by illuminating the material with two mutually coherent laser beams. During the recording process, the monomers polymerize, and the mixture undergoes a photopolymerization-induced phase separation, creating regions densely populated by liquid crystal (LC) micro-droplets, interspersed with regions of clear polymer. The alternating liquid crystal-rich and liquid crystal-depleted regions form the fringe planes of the grating.

Waveguide optics, such as those described above, can be considered for a range of display and sensor applications. In many applications, waveguides containing one or more grating layers encoding multiple optical functions can be realized using various waveguide architectures and material systems, enabling new innovations in near-eye displays for Augmented Reality (“AR”) and Virtual Reality (“VR”), compact Heads Up Displays (“HUDs”) for aviation and road transport, and sensors for biometric and laser radar (“LIDAR”) applications. As many of these applications are directed at consumer products, there is a growing requirement for efficient low cost means for manufacturing holographic waveguides in large volumes.

BRIEF DESCRIPTION OF THE DRAWINGS

The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application

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publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

The description will be more fully understood with reference to the following figures and data graphs, which are presented as various embodiment of the disclosure and should not be construed as a complete recitation of the scope of the disclosure, wherein:

FIG. 1A illustrates a profile view of a waveguide-based display in accordance with an embodiment of the invention.

FIG. 1B shows a plan view of the waveguide shown in FIG. 1A.

FIG. 1C shows a profile view of a waveguide-based display in accordance with an embodiment of the invention.

FIG. 2A illustrates a portion of a surface relief grating in accordance with an embodiment of the invention.

FIG. 2B illustrates a cross-sectional view of a photonic crystal in accordance with an embodiment of the invention.

FIG. 3 conceptually illustrates a cross-sectional view of a waveguide-based display in accordance with an embodiment of the invention.

FIG. 4 conceptually illustrates a cross-sectional view of a waveguide-based display in accordance with an embodiment of the invention.

FIG. 5 conceptually illustrates a plan view of the waveguide-based display described in connection with FIG. 3.

FIG. 6 conceptually illustrates in cross section a waveguide-based display in accordance with an embodiment of the invention.

FIG. 7 conceptually illustrates a plan view of a waveguide-based display in accordance with an embodiment of the invention.

FIGS. 8A-8D illustrate a processing apparatus that can be used in a method for fabricating deep SRGs or EBGs in accordance with an embodiment.

FIG. 9 conceptually illustrates a method for forming a surface relief grating from a HPDLC Bragg grating formed on a transparent substrate in accordance with an embodiment of the invention.

FIG. 10A conceptually illustrates an apparatus that can be used in a step of a method for fabricating a hybrid surface relief/Bragg grating in accordance with an embodiment of the invention.

FIG. 10B conceptually illustrates an apparatus that can be used in a step of a method for fabricating a surface relief grating in accordance with an embodiment of the invention.

FIG. 10C conceptually illustrates an apparatus that can be used in a step of a method for fabricating a surface relief grating in accordance with an embodiment of the invention.

FIG. 10D conceptually illustrates an apparatus that can be used in a step of a method for fabricating a surface relief grating in which the surface relief grating formed in the step illustrated in FIG. 10C is partially refilled with liquid crystal in accordance with an embodiment of the invention.

FIG. 10E conceptually illustrates an apparatus that can be used in a step of a method for fabricating a surface relief grating in which the hybrid surface relief Bragg grating formed in the step illustrated in FIG. 10D is covered with a protective layer, in accordance with an embodiment of the invention.

FIG. 11 is a flowchart showing an exemplary method for forming a hybrid surface relief-Bragg grating from a HPDLC Bragg grating formed on a transparent substrate in accordance with an embodiment of the invention.

FIG. 12 is a graph showing calculated P-polarized and S-polarized diffraction efficiency versus incidence angle for a 1-micron thickness deep surface relief grating demonstrating that in this case high S and P response can be achieved.

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FIG. 13 is a graph showing calculated P-polarized and S-polarized diffraction efficiency versus incidence angle for a 2-micron thickness deep surface relief grating, demonstrating that in this case the S-polarization response is dominant over most of the angular range of the grating.

FIG. 14 is a graph showing calculated P-polarized and S-polarized diffraction efficiency versus incidence angle for a 3-micron thickness, demonstrating that in this case the P-polarization response is dominant over a substantial portion of the angular range of the grating.

FIGS. 15A-15B shows a plot of calculated angular and spectral diffraction efficiency characteristics for a reflection structure formed from a HPDLC in accordance with an embodiment of the invention.

FIGS. 16A-16B shows the corresponding diffraction efficiency characteristics of a grating formed from a polymer of index 1.8 (refractive index modulation 0.4) for the same beam angles in accordance with an embodiment of the invention.

FIGS. 17A-17E illustrate the step in fabricating such a reflection Bragg grating incorporating polymer scaffolding in accordance with an embodiment of the invention.

FIG. 18 conceptually illustrates a reflection grating in accordance with an embodiment of the invention.

FIG. 19 conceptually illustrates a reflection grating in accordance with an embodiment of the invention.

FIG. 20 conceptually illustrates a combination reflection and transmission grating in accordance with an embodiment of the invention.

FIG. 21 conceptually illustrates a waveguide-based display in accordance with an embodiment of the invention.

FIG. 22 illustrates a plot of different materials that have different dispersion (n-A) curves which overlap at two or more points.

FIG. 23 conceptually illustrates a portion of a Bragg grating formed from different dispersion materials in accordance with an embodiment of the invention.

FIGS. 24-27 conceptually illustrate waveguide gratings with spatially varying diffraction efficiency in accordance with an embodiment of the invention.

FIG. 28 conceptually illustrates a waveguide apparatus in accordance with an embodiment of the invention.

FIG. 29 conceptually illustrates a waveguide apparatus in accordance with an embodiment of the invention.

FIG. 30 conceptually illustrates a waveguide apparatus in accordance with an embodiment of the invention.

FIG. 31 conceptually illustrates a portion of a grating structure in accordance with an embodiment of the invention.

FIG. 32 conceptually illustrates a portion of a grating structure in accordance with an embodiment of the invention.

FIG. 33 conceptually illustrates a portion of a grating structure in accordance with an embodiment of the invention.

FIG. 34 conceptually illustrates a waveguide apparatus in accordance with an embodiment of the invention.

FIG. 35 conceptually illustrates a waveguide apparatus in accordance with an embodiment of the invention.

FIG. 36A conceptually illustrates a waveguide-based display in accordance with an embodiment of the invention.

FIG. 36B conceptually illustrates a waveguide-based display in accordance with an embodiment of the invention.

FIG. 36C conceptually illustrates a waveguide-based display in accordance with an embodiment of the invention.

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FIG. 37 conceptually illustrates a side elevation view of a waveguide-based display based on the embodiments of FIGS. 36A, 36B, 36C.

FIG. 38 illustrates a plan view of the waveguide-based display described in connection with FIG. 37.

FIG. 39 conceptually illustrates an apparatus for recording a grating structure including a grating array in accordance with an embodiment of the invention.

FIG. 40 illustrates a multiplexed grating structure in accordance with an embodiment of the invention.

FIG. 41A conceptually illustrates a portion of a grating element pattern including rectangular elements of differing size and aspect ratio for use in an emissive display panel in accordance with an embodiment of the invention.

FIG. 41B conceptually illustrates a portion of a grating element pattern including Penrose tiles in accordance with an embodiment of the invention.

FIG. 41C conceptually illustrates a portion of a grating element pattern including hexagons in accordance with an embodiment of the invention.

FIG. 41D conceptually illustrates a portion of a grating element pattern including squares in accordance with an embodiment of the invention.

FIG. 41E conceptually illustrates a portion of a grating element pattern comprising diamond-shaped elements in accordance with an embodiment of the invention.

FIG. 41F conceptually illustrates a portion of a grating element pattern including isosceles triangles in accordance with an embodiment of the invention.

FIG. 41G conceptually illustrates a portion of a grating element pattern including hexagons of horizontally biased aspect ratio in one embodiment.

FIG. 41H conceptually illustrates a portion of a grating element pattern including rectangles of horizontally biased aspect ratio in accordance with an embodiment of the invention.

FIG. 41I conceptually illustrates a portion of a grating element pattern including diamond shaped elements of horizontally biased aspect ratio in accordance with an embodiment of the invention.

FIG. 41J conceptually illustrates a portion of a grating element pattern including triangles of horizontally biased aspect ratio in accordance with an embodiment of the invention.

FIG. 42 conceptually illustrates a method for forming a waveguide color grating structure in accordance with an embodiment of the invention.

SUMMARY OF THE DISCLOSURE

In many of the embodiments to be described there is provided A waveguide-based display device including: a waveguide; a source of image modulated light projected over a field of view; an input coupler for coupling said light into a total internal reflection (TIR) path within the waveguide; and a grating structure for providing beam expansion of the TIR light in at least one direction and extracting the TIR light from the waveguide, wherein the grating structure comprises a plurality of grating elements having at least two different grating prescriptions and at least one surface relief grating formed by a phase separation process.

In many embodiments, the plurality of grating elements are tiled in a repeating pattern. In many embodiments, the plurality of grating elements are tiled in a non-repeating pattern.

In many embodiments, the grating structure further includes a spatial refractive index modulation.

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In many embodiments, the plurality of grating elements includes at least two different spectral responses.

In many embodiments, the plurality of grating elements includes at least two different angular responses.

In many embodiments, the plurality of grating elements includes at least two different birefringence characteristics.

In many embodiments, the plurality of grating elements includes at least two different polarization responses.

In many embodiments, the plurality of grating elements includes at least two different refractive index modulations.

In many embodiments, the plurality of grating elements includes an average refractive index including at least two different refractive indices.

In many embodiments, the plurality of grating elements includes at least two different grating thicknesses.

In many embodiments, the grating structure can comprise a plurality of grating elements having grating prescriptions selected from a group containing at least two different grating prescriptions.

In many embodiments, the plurality of grating elements includes at least two different K-vectors.

In many embodiments, the plurality of grating elements includes diffracting grating elements arrayed with non-diffracting grating elements.

In many embodiments, the plurality of grating elements is recorded in material containing at least two different materials.

In many embodiments, the plurality of grating elements is recorded in material having dispersion curves selected from a group containing at least two different dispersion curves.

In many embodiments, the plurality of grating elements is recorded in material having at least two different wavelength sensitivities.

In many embodiments, the plurality of grating elements is recorded in material having at least two different holographic exposure times.

In many embodiments, the plurality of grating elements is recorded in material having at least two different holographic exposure energies.

In many embodiments, the grating structure is configured to enable a plurality of ray path lengths within the grating structure that differ in length by a distance shorter than the coherence length of the source of image modulated light.

In many embodiments, the plurality of grating elements includes at least one grating pitch greater than a wavelength of the TIR light.

In many embodiments, the plurality of grating elements includes a spatial variation of at least one selected from the group consisting of: spatial variation of grating thickness, average refractive index, refractive index modulation, and birefringence.

In many embodiments, the grating structure includes at least one selected from a group consisting of: a rolled K-vector grating, a dual interaction grating, a multiplexed grating, a lossy grating, a sub wavelength grating, and a chirped grating.

In many embodiments, the at least one surface relief grating is formed by liquid crystal extraction from a grating recorded in holographic polymer dispersed liquid crystal (HPDLC).

In many embodiments, the at least one surface relief grating is formed by at least partially backfilling a surface relief grating formed by liquid crystal extraction from a holographic polymer dispersed liquid crystal (HPDLC) recorded grating with another material having a refractive index higher than that of the extracted liquid crystal.

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In many embodiments, the at least one surface relief grating is formed by at least partially backfilling a surface relief grating formed by liquid crystal extraction from a holographic polymer dispersed liquid crystal (HPDLC) recorded grating with another material having a refractive index lower than that of the extracted liquid crystal.

In many embodiments, the grating structure includes at least one reflection grating. In many embodiments, the grating structure includes at least one transmission grating.

In many embodiments, the grating structure can be a switching grating.

In many embodiments, the grating structure can further comprise at least one selected from the group of a beam-splitter layer, an anti-reflection coating, an optical bandpass filter, a polarization modification layer, and/or an alignment layer.

In many embodiments, the grating structure provides beam expansion and extraction of red, green and blue image modulated light over a field of view. In many embodiments, the image modulated light is monochromatic.

In many embodiments, the grating structure is recorded into a material selected from the group of a mixture including at least one liquid crystal and at least one monomer, a mixture including at least one blue phase liquid crystal and at least one monomer.

In many embodiments, the input coupler can be a grating or a prism. In many embodiments the input coupler can include a plurality of gratings. In many embodiments, the input coupler can be a switching grating.

In many embodiments, the plurality of grating elements are spatially distributed elements.

In many embodiments, the at least one surface relief grating includes a grating modulation depth greater than a grating pitch.

In many of the embodiments to be described a waveguide can support at least one photonic crystal. For the purposes of explaining the invention a photonic crystal can be considered as a periodic optical nanostructure that affects the motion of photons. Photonic crystals can be fabricated for one, two, or three dimensions. An example of a one-dimensional photonic crystal is a grating structure formed from alternating layers of high refractive index and low refractive index materials. Such gratings are commonly referred to as Bragg or volume gratings. A two-dimensional photonic crystal is formed by a two-dimensional array of elements of a first refractive index immersed in a material of a second refractive index. Two-dimensional photonic crystals can be fabricated by photolithography, or by drilling holes in a suitable substrate. Fabrication methods for three-dimensional photonic crystal include drilling under different angles, stacking multiple 2-D layers on top of each other and direct laser writing. Another approach is forming a matrix of spheres or instigating self-assembly of spheres in a matrix and dissolving the sphere. In many cases, the regions of low refractive index are provided by air. Two-dimensional photonic crystals can be any of the 5 2-D Bravais lattices. Typically, photonic crystals have periodicity of around half the wavelength of the light to be diffracted. The low dielectric can be provided by air. Three-dimensional photonic crystals can provide any of the fourteen 3D Bravais lattices.

In many of the embodiments of the invention to be described below, a photonic crystal including a grating structure immersed at least partially in air can be formed from a mixture of liquid crystal and monomer materials using a phase separation process taking place under holographic exposure. After exposure is complete liquid crystal can be removed from the surface relief grating. In many

embodiments, the grating structure can be refilled with a different material such as a LC of a different index and different other properties. In many embodiments, the grating structure can be partially backfilled to provide a hybrid surface relief and volume grating structure. In many embodiments, the grating structure can be refilled with an organic or inorganic material with a high refractive index. In many embodiments, the grating can have material properties varying spatially can varying depth. In many embodiments, the backfilling can be a diffusion process. In many embodiments, the grating structure can be backfilled with chemical components that are phase separated under a laser exposure process. In many embodiments, backfilling can be carried out in the presence of thermal, mechanical, chemical or electromagnetic stimuli for influencing annealing and alignment of the grating structure.

In some embodiments, the diffractive surface can be a metasurface, where for the purposes of discussing the invention a metasurface is defined as a surface structure supporting subwavelength in-plane features that are used to realize a desirable functionality by locally managing the interaction between the metasurface and incident electromagnetic fields. Metasurfaces are typically formed on sub-wavelength thickness substrates. However, much thicker substrates may be used. Metasurfaces may have diffractive features of nanometre-scale. This enables a much greater degree of wavefront phase and amplitude control than can be achieved using conventional diffractive optical elements (DOEs). In many applications, diffraction limited performance is a realistic goal for optical components (including lenses) made using metasurfaces. Conventional DOEs operate at the micron scale with feature heights that are significant fractions of visible band wavelengths, resulting in more limited capacity for wavefront amplitude and phase control, especially when manufacturing tolerances are taken into account.

In many embodiments, a photonic crystal provides an input grating. As will be discussed in the following paragraphs a photonic crystal formed by liquid crystal extraction offers potential benefits in terms of improving the angular bandwidth of a waveguide and can be used to control the polarization characteristics of waveguided light. The various embodiments to be discussed can be applied in HUDs for automotive applications, near eye displays and other waveguide display applications. In many embodiments, photonic crystal according to the principles of the invention can be used in single axis or in dual expansion waveguides. In some embodiments, photonic crystals can be used to provide beam expansion gratings. In some embodiments, photonic crystals can be used to provide output gratings. In some embodiments, photonic crystals can be used to diffract more than one primary color. In some embodiments, waveguides incorporating photonic crystals can be arranged in stacks of waveguides each having grating prescription for diffracting a unique spectral bandwidth.

DETAILED DESCRIPTION

FIG. 1A illustrates a profile view of a waveguide-based display in accordance with an embodiment of the invention. The waveguide-based display **100** includes a waveguide **102** including an input coupler **104** and an output coupler **106**. Image containing light is incoupled by the input coupler **104** into total internal reflection (TIR) in the waveguide **102**. The TIR light is then outcoupled by the output coupler **106** into outcoupled light **110**. FIG. 1B shows a plan view of the waveguide shown in FIG. 1A.

FIG. 1C shows a profile view of a waveguide-based display in accordance with an embodiment of the invention. The waveguide-based display **100a** shares many components identically labelled with the waveguide-based display **100** of FIGS. 1A and 1B. The description of those components is applicable here and will not be repeated in detail. The waveguide-based display **100a** includes an input prism **104a** which operates to incouple image containing light **108** into TIR in the waveguide **102**.

FIG. 2A illustrates a portion of a surface relief grating in accordance with an embodiment of the invention. The surface relief grating **110A** includes an optical substrate **101** supporting grating elements **102** separated by air gaps **103**.

FIG. 2B illustrates a cross-sectional view of a photonic crystal in accordance with an embodiment of the invention. The photonic crystal **110B** includes a three-dimensional lattice elements including grating features **111** separated by air spaces **112**. In some embodiments, the lattice elements can include air regions surround by an optical material. In some embodiments, the lattice can be one of the Bravais lattices. Photonic crystals are described in detail in PCT App. No. PCT/US2021/072548, entitled "Photonic Crystals and Methods for Fabricating the Same" and filed on Nov. 22, 2021, which is hereby incorporated by reference in its entirety.

FIG. 3 conceptually illustrates a cross-sectional view of a waveguide-based display in accordance with an embodiment of the invention. The waveguide-based display **120** includes a waveguide **121** including a reflective input grating **122** and an output grating **123**. Input light **124** from a picture generation unit (PGU) is coupled into the waveguide by the input grating **122** and propagates along a total internal reflection path indicated by the rays **125-127** before being extracted by the output grating **123** which provides one dimensional beam expansion into output light **128**. In some embodiments, the waveguide further supports a fold grating.

FIG. 4 conceptually illustrates a cross-sectional view of a waveguide-based display in accordance with an embodiment of the invention. The waveguide-based display **130** includes a waveguide **131** supporting an input grating **132** and an output grating **133**. Input light **134** from a picture generation unit (PGU) is coupled into the waveguide **131** by the input grating **132** and propagates along a total internal reflection path indicated by the rays **135-136** before being extracted by the output grating **133** which provides one dimensional beam expansion. In many embodiments, at least one of the gratings is a photonic crystal formed using a liquid crystal extraction process.

FIG. 5 conceptually illustrates a plan view of the waveguide-based display **120** described in connection with FIG. 3. FIG. 5 further shows the PGU **141** which outputs image containing light **124** into the input grating **122**.

FIG. 6 conceptually illustrates in cross section a waveguide-based display in accordance with an embodiment of the invention. The waveguide-based display **150** includes a waveguide **151** supporting a reflection input grating **122**, a fold grating **152** which provides a first direction beam expansion, and an output grating **153**. Input light **145** from a PGU **141** is coupled into the waveguide **151** by the input grating **122** and propagates along total internal reflection paths indicated by the rays **154,155** before being extracted **156** by the output grating **153** which provides a second beam expansion orthogonal to the first beam expansion. Rays in FIGS. 5-7 (such as rays **154,155**) are in TIR between the reflection faces of the waveguide. The ray bounces illustrated would only be seen in the waveguide cross section.

In many embodiments, at least one of the gratings **122**, **152**, **153** is a photonic crystal formed using a liquid crystal extraction process.

FIG. 7 conceptually illustrates a plan view of a waveguide-based display in accordance with an embodiment of the invention. The waveguide-based display **160** includes a waveguide **161** which supports an input grating **162**, and a pair of overlapping or multiplexed fold gratings **163**, **164**. A PGU **141** outputs image containing light **145** into the input grating **162**. The multiplexed fold gratings **163**, **164** provide beam expansion in two orthogonal dimensions. The total internal reflection path from the input grating **162** to the fold gratings **163**, **164** and the pupil-expanded light extracted from the waveguide **161** by the overlapping fold gratings **163**, **164** are represented by the rays **165**, **166**. In many embodiments at least one of the gratings **162**, **163**, **164** is a photonic crystal formed using a liquid crystal extraction process.

In some embodiments, LC can be extracted or evacuated from the SBG to provide an evacuated periodic grating (EPG). EPGs can be characterized as a surface relief grating (SRG) that has properties very similar to a Bragg grating due to the depth of the SRG structure (which is much greater than that practically achievable using surface etching and other conventional processes commonly used to fabricate SRGs). The LC can be extracted using a variety of different methods, including but not limited to flushing with isopropyl alcohol and solvents. In many embodiments, one of the transparent substrates of the SBG is removed, and the LC is extracted. In further embodiments, the removed substrate is replaced. The SRG can be at least partially backfilled with a material of higher or lower refractive index. Such gratings offer scope for tailoring the efficiency, angular/spectral response, polarization, and other properties to suit various waveguide applications. Examples of EPGs and methods for manufacturing EPGs are discussed in US Pat. Pub. No. 2021/0063634, entitled "Evacuating Bragg Gratings and Methods of Manufacturing" and filed Aug. 28, 2020 which is hereby incorporated by reference in its entirety for all purposes.

In many embodiments a photonic crystal formed by liquid crystal extraction provide an EPG which may be a deep surface relief grating (SRG). In many embodiments, a deep SRG formed using a liquid crystal extraction process can typically have a thickness in the range 1-3 micron with a Bragg fringe spacing 0.35 micron to 0.80 micron. In many embodiments, the condition for a deep SRG is characterized by a high grating depth to fringe spacing ratio. In some embodiments, the condition for the formation of a deep SRG is that the grating depth can be approximately twice the grating period. Modelling such SRGs using the Kogelnik theory can give reasonably accurate estimates of diffraction efficiency avoiding the need for more advanced modelling which typically entails the numerical solution of Maxwell's equations. The grating depths that can be achieved using liquid crystal removal from HPDLC gratings greatly surpass those possible using conventional nanoimprint lithographic methods, which do not achieve the condition for a deep SRG, typically providing only 250-300 nm. depth for grating periods 350-460 nm. (Pekka Äyräs, Pasi Saarikko, Tapani Levola, "Exit pupil expander with a large field of view based on diffractive optics", Journal of the SID 17/8, (2009), pp 659-664). Although deep SRGs may diffract S-polarization, deep SRGs can provide a range of polarization response characteristics depending on the thickness of the grating prescription and, in particular, the grating depth. Deep SRGs can also be used in conjunction with conven-

tional Bragg gratings to enhance the color, uniformity and other properties of waveguide displays.

SRGs have been fabricated in glassy monomeric azobenzene materials using laser holographic exposure (O. Sakhno, L. M. Goldenberg, M. Wegener, J. Stumpe, "Deep surface relief grating in azobenzene-containing materials using a low intensity 532 nm laser", Optical Materials: X, 1, (2019), 100006, pp 3-7. The Sakhno reference also discloses how SRGs can be recorded in a holographic photopolymer using two linearly orthogonally polarized laser beams.

Deep SRG and Bragg gratings are also disclosed in Kiyoshi Yokomori, "Dielectric surface-relief gratings with high diffraction efficiency" Applied Optics; Vol. 23; Issue 14; (1984); pp. 2303-2310 which discloses the investigation of the diffraction properties of dielectric surface-relief gratings by solving Maxwell's equations numerically. The diffraction efficiency of a grating with a groove depth about twice as deep as the grating period was found to be comparable with the efficiency of a volume phase grating. The modelling by Yokomori predicted that dielectric surface-relief gratings interferometrically recorded in photoresist can possess a high diffraction efficiency of up to 94% (throughput efficiency 85%). The equivalence of deep SRGs and Bragg gratings is also discussed in another article by Golub (M. A. Golub, A. A. Friesem, L. Eisen "Bragg properties of efficient surface relief gratings in the resonance domain", Optics Communications; 235; (2004); pp 261-267). A further article by Gerritsen discusses the formation of Bragg-like SRGs in photoresist (Gerritsen H J, Thornton D K, Bolton S R; "Application of Kogelnik's two-wave theory to deep, slanted, highly efficient, relief transmission gratings" Applied Optics; Vol. 30; Issue 7; (1991); pp 807-814).

The invention provides a method for making a surface relief grating that can offer very significant advantages over nanoimprint lithographic process particle for slanted gratings. Bragg gratings of any complexity can be made using interference or master and contact copy replication. In some embodiments after removing the LC the SRG can be back filled with a material with different properties to the LC. This allows a Bragg grating with modulation properties that are not limited by the grating chemistry needed for grating formation. In some embodiments the deep SRG can be partially backfilled with LC to provide a hybrid SRG/Bragg grating. Alternatively, in some embodiments, the refill step can be avoided by removing just a portion of the LC from the LC rich regions of the HPDLC to provide a hybrid SRG/Bragg grating. The refill approach has the advantage that a different LC can be used to form the hybrid grating. In some embodiments, the backfill materials can be deposited using inkjet process.

FIGS. 8A-8D illustrate a processing apparatus that can be used in a method for fabricating deep SRGs or EBGs in accordance with an embodiment. FIG. 8A conceptually illustrates an apparatus **170A** that can be used in a step of a method for fabricating a surface relief grating in which a mixture **171** of monomer and liquid crystal deposited on a transparent substrate **172** is exposed to holographic exposure beams **173**, **174**, in accordance with an embodiment of the invention. In some examples, the mixture may also include at least one of a photoinitiator, a coinitiator, a multifunctional thiol, adhesion promoter, surfactant, and/or additional additives. In some embodiments, the monomer may be isocyanate-acrylate based or thiolene based. In some embodiments, the liquid crystal may be a full liquid crystal mixture or a liquid crystal single only including a portion of a full liquid crystal mixture. Various examples of liquid

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crystal singles include one or both of cyanobiphenyls or pentylycyanobiphenyls. In some embodiments, liquid crystal may be replaced with another substance that phase separates with the monomer during exposure to create polymer rich regions and substance rich regions. Advantageously, the substance and liquid crystal singles may be a cost-effective substitute to full liquid crystal mixtures which are removed at a later step as described below.

FIG. 8B conceptually illustrates an apparatus 170B that can be used in a step of a method for fabricating a surface relief grating from an HPDLC Bragg grating 175 formed on a transparent substrate using the holographic exposure beams, in accordance with an embodiment of the invention. The holographic exposure beams may transform the monomer into a polymer in some areas. The holographic exposure beams may include intersecting recording beams and include alternating bright and dark illumination regions. A polymerization-driven diffusion process may cause the diffusion of monomers and LC in opposite directions, with the monomers undergoing gelation to form polymer-rich regions (in the bright regions) and the liquid crystal becoming trapped in a polymer matrix to form liquid crystal rich regions (in the dark regions).

FIG. 8C conceptually illustrates an apparatus 170C that can be used in a step of a method for fabricating a deep polymer surface relief grating 176 or EBG in which liquid crystal is removed from an HPDLC Bragg grating of FIG. 1B to form a polymer surface relief grating in accordance with an embodiment of the invention. Advantageously, a polymer surface relief grating 176 may include a large depth with a comparatively small grating period in order to form a deep SRG. The liquid crystal may be removed by washing with a solvent such as isopropyl alcohol (IPA). The solvent should be strong enough to wash away the liquid crystal but weak enough to maintain the polymer. In some embodiments, the solvent may be chilled below room temperature before washing the grating. FIG. 8D conceptually illustrates an apparatus 170D that can be used in a step of a method for fabricating a polymer surface relief grating in which the polymer surface relief grating is covered with a protective layer 177 in accordance with an embodiment of the invention.

FIG. 9 conceptually illustrates a method for forming a surface relief grating from a HPDLC Bragg grating formed on a transparent substrate in accordance with an embodiment of the invention. As shown, the method 180 of forming a surface relief grating is provided. Referring to the flow diagram, method 180 includes providing (181) a mixture of at least one monomer and at least one liquid crystal. A transparent substrate can be provided (182). A layer of the mixture can be deposited (183) onto a surface of the substrate. Holographic recording beams can be applied (184) to the mixture layer. A HPDLC grating comprising alternating polymer rich and liquid crystal rich regions can be formed (185). The liquid crystal in the liquid crystal rich regions can be removed (186) to form a polymer surface relief grating.

As discussed above, in many the embodiments, the invention also provides a method for fabricating a hybrid surface relief/Bragg grating. FIG. 10A conceptually illustrates apparatus 190A that can be used in a step of a method for fabricating a hybrid surface relief/Bragg grating in which a mixture 191 of monomer and liquid crystal deposited on a transparent substrate 212 is exposed to holographic exposure beams 193, 194, in accordance with an embodiment of the invention. FIG. 10B conceptually illustrates apparatus 190B that can be used in a step of a method for fabricating a

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surface relief grating from a HPDLC Bragg grating 195 formed on the transparent substrate using the holographic exposure beams in accordance with an embodiment of the invention. FIG. 10C conceptually illustrates apparatus 190C that can be used in a step of a method for fabricating a surface relief grating in which liquid crystal is removed from the HPDLC Bragg grating to form a surface relief grating 196 in accordance with an embodiment of the invention. FIG. 10D conceptually illustrates apparatus 190D that can be used in a step of a method for fabricating a surface relief grating in which the surface relief grating formed in the step illustrated in FIG. 10C is partially refilled with liquid crystal to form a hybrid surface relief/Bragg grating 197 in accordance with an embodiment of the invention. FIG. 10E conceptually illustrates apparatus 190E that can be used in a step of a method for fabricating a surface relief grating in which the hybrid surface relief Bragg grating formed in the step illustrated in FIG. 10D is covered with a protective layer 198, in accordance with an embodiment of the invention.

FIG. 11 is a flowchart showing an exemplary method for forming a hybrid surface relief-Bragg grating from a HPDLC Bragg grating formed on a transparent substrate in accordance with an embodiment of the invention. As shown, the method 220 of forming hybrid surface relief-Bragg grating is provided. Referring to the flow diagram, method 220 includes providing (221) a mixture of at least one monomer and at least one liquid crystal. The at least one monomer may include an isocyanate-acrylate monomer. Providing the mixture of the monomer and the liquid crystal may also include mixing one or more of the following with the at least one monomer and the liquid crystal: photoinitiator, coinitiator, multifunctional thiol, and/or additional additives. This mixture may be allowed to rest in order to allow the coinitiator to catalyze a reaction between the monomer and the thiol. The rest period may occur in a dark space or a space with red light (e.g. infrared light) at a cold temperature (e.g. 20° C.) for a period of approximately 8 hours. After resting, additional monomers may be mixed into the monomer. This mixture may be then strained or filtered through a filter with a small pore size (e.g. 0.45 µm pore size). After straining this mixture may be stored at room temperature in a dark space or a space with red light before coating.

Next, a transparent substrate can be provided (222). In certain embodiments, the transparent substrate may be a glass substrate or a plastic substrate. A non-stick coating may be applied to the transparent substrate before the mixture is coated on the substrate. A layer of the mixture can be deposited (223) onto a surface of the substrate. In some embodiments, the mixture is sandwiched between the transparent substrate and another substrate using glass spacers to maintain internal dimensions. Holographic recording beams can be applied (224) to the mixture layer. The holographic recording beams may be a two-beam interference pattern which may cause phase separation of the LC and the polymer. After applying the holographic recording beams, the mixture may be cured. The curing process may include leaving the mixture under low-intensity white light for a period of time under the mixture fully cures. The low intensity white light may also cause a photo-bleach dye process to occur. Thus, an HPDLC grating having alternating polymer rich and liquid crystal rich regions can be formed (225). In some embodiments, the curing process may occur in 2 hours or less. After curing, one of the substrates may be removed exposing the HPDLC grating.

HPDLC grating may include alternating sections of liquid crystal rich regions and polymer regions. The liquid crystal in the liquid crystal rich regions can be removed (226) to form polymer surface relief gratings or EBGs which is a form of deep SRGs. The liquid crystal may be removed by gently immersing the grating into a solvent such as isopropyl alcohol (IPA). The IPA may be kept at a lower temperature while the grating is immersed in the IPA. The grating is then removed from the solvent and dried. In some embodiments, the grating is dried using a high flow air source such as compressed air. After the LC is removed from the grating, a polymer-air surface relief Bragg grating is formed. The steps 221-226 of FIG. 11 roughly correspond to the steps described in connection with FIG. 10 in creating a polymer-air SRG and thus these descriptions are applicable to FIG. 11.

Further, method 220 includes at least partially refilling (227) cleared liquid crystal rich regions with liquid crystal to form hybrid SRGs. The refilled liquid crystal may be of different consistency to the previously removed liquid crystal that was previously removed in step 226. Further, it is appreciated that the liquid crystal removed in step 226 may only be partially removed in an alternative method to forming hybrid SRGs. Advantageously, hybrid SRGs may provide the ability to tailor specific beneficial characteristics of the SRGs. One particular characteristic that may be improved by the inclusion of at least some liquid crystal within the SRGs is a decrease in haze properties.

As shown in FIG. 10E, the formed surface relief grating can further be covered with a protective layer. In some instances, the protective layer may be a moisture and oxygen barrier with scratch resistance capabilities. In some instances, the protective layer may be a coating that does not fill in air gap regions where LC that was removed once existed. The coating may be deposited using a low temperature process. In some implementations, the protective layer may have anti-reflective (AR) properties. The coating may be a silicate or silicon nitride. The coating process may be performed by a plasma assisted chemical vapor deposition (CVD) process such as a plasmatreat nanocoating process. The coating may be a parylene coating. The protective layer may be a glass layer. A vacuum or inert gas may fill the gaps where LC that was removed once existed before the protective layer is implemented. In some embodiments, the coating process may be integrated with the LC removal process (226). For example, a coating material may be mixed with the solvent which is used to wash the LC from the grating. In some implementations, the coating material may be a material with a lower or higher refractive index than the polymer and fill the spaces between adjacent polymer portions. The refractive index difference between the polymer and the coating material may allow the polymer SRGs to continue to diffract. Hybrid SRG/Bragg gratings suffer from the shallowness of the SRG structure which leads to low SRG diffraction efficiencies. The method disclosed in the present application allows more effective SRG structures to be formed by optimised the depth of the liquid crystal in the liquid crystal rich regions such that the SRG has a high depth to grating pitch ratio while allowing the Bragg grating to be sufficiently thick for efficient diffraction. In many embodiments the Bragg grating component of the hybrid grating can have a thickness in the range 1-3 micron. In many embodiments, the SRG component of the hybrid

sum of the final SRG and Bragg grating components. The thickness ratio of the two grating components depends on the waveguide application.

In many embodiments, the refill depth of the liquid crystal regions of the grating can be varied across the grating to provides spatially varying relative SRG/Bragg grating strengths. In many embodiments, as an alternative to liquid crystal removal and refill, the liquid crystal in the liquid crystal rich grating regions can be totally or partially removed. In many embodiments, the liquid crystal used to refill or partially refill the liquid crystal-cleared regions can have a different chemical composition to the liquid crystal used to form the HPDLC grating. In some embodiments, a first liquid crystal with phase separation properties compatible with the monomer can be specified to provide a HPDLC grating with optimal modulation and grating definitions while a second refill liquid crystal can be specified to provide desired index modulation properties in the final hybrid grating. In many embodiments, the Bragg portion of the hybrid grating can be switchable with electrodes applied to surfaces of the substrate and the cover layer. In many embodiments the refill liquid crystals can contain additives for improving switching voltage, switching time, polarization, transparency, and other parameters. A hybrid grating formed using a refill process would have the further advantages that the LC would be form a continuum (rather than an assembly of LC droplets) thereby reducing haze.

In many embodiments, a deep SRG can provide a means for controlling polarization in a waveguide. SBGs are normally P-polarization selective leading to a 50% efficiency loss with unpolarized light sources such as OLEDs and LEDs. Hence combining S-polarization diffracting and P-polarization diffracting gratings as discussed above can provide a theoretical 2x improvement over waveguides using P-diffracting gratings only. In some embodiments a S-polarization diffracting grating can be provided by a Bragg grating formed in a conventional holographic photopolymer. In some embodiments an S-polarization diffracting grating can be provided by a Bragg grating formed in a HPDLC with birefringence altered using an alignment layer or other process for realigning the liquid crystal directors. In some embodiments an S-polarization diffracting grating can be formed using liquid crystals, monomers and other additives that naturally organize into S-diffracting gratings under phase separation. In many embodiments, an S-polarization diffracting grating can be provided by a surface relief grating (SRG). The inventors have discovered by experimentation that a deep SRG exhibiting high S-diffraction efficiency (up to 99%) and low P-diffraction efficiency can be formed by removing the liquid crystal from an SBG formed from holographic phase separation of a liquid crystal and monomer mixture.

Deep SRGs can also provide other polarization response characteristics. Several prior art theoretical studies such as an article by Moharam (Moharam M. G. et al. "Diffraction characteristics of photoresist surface-relief gratings", Applied Optics, Vol. 23, page 3214, Sep. 15, 1984) point to deep surface relief gratings having both S and P sensitivity with S being dominant. In many embodiments, the thickness of the SRG can be adjusted to provide a variety of S and P diffraction characteristics. In some embodiments, diffraction efficiency can be high for P across a spectral bandwidth and angular bandwidth and low for S across the same spectral bandwidth and angular bandwidth. In some embodiments, diffraction efficiency can be high for S across the spectral bandwidth and angular bandwidth and low for P across the same spectral bandwidth and angular bandwidth. In some

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embodiments, high efficiency for both S and P polarized light can be provided. Analysis of a SRG of refractive index 1.6 immersed in air (hence providing an average grating index of 1.3) of period 0.48 micron, with a 0 degrees incidence angle and 45 degree diffracted angle for a wave-length of 0.532 micron have been performed. FIG. 12 is a graph 210 showing calculated P-polarized and S-polarized diffraction efficiency versus incidence angle for a 1-micron thickness deep surface relief grating demonstrating that in this case high S and P response can be achieved. FIG. 13 is a graph 220 showing calculated P-polarized and S-polarized diffraction efficiency versus incidence angle for a 2-micron thickness deep surface relief grating, demonstrating that in this case the S-polarization response is dominant over most of the angular range of the grating. FIG. 14 is a graph 230 showing calculated P-polarized and S-polarized diffraction efficiency versus incidence angle for a 3-micron thickness, demonstrating that in this case the P-polarization response is dominant over a substantial portion of the angular range of the grating.

In many embodiments, the photonic crystal can be a reflection Bragg grating formed by a LC extraction process. A reflection Bragg grating made using phase separation followed by LC subtraction can enable wide angular and spectral bandwidth. In many embodiments replacing the current input SBG with a reflection photonic crystal can be used to reduce the optical path from the PGU to the waveguide. In some embodiments, the PGU pupil and the waveguide can be in contact. In many embodiments, the reflection Bragg grating can be approximately 3 microns in thickness. The diffracting properties of an LC extracted Bragg grating mainly result from the index gap between the polymer and air (not from the depth of the grating as in the case of a SRG).

FIGS. 15A-15B shows a plot of calculated angular and spectral diffraction efficiency characteristics for a reflection structure formed from a HPDLC in accordance with an embodiment of the invention. The input and diffracted beam angles are 0° and 45° and the grating thickness is 3 microns. The refractive index of the polymer component of the grating is 1.6 and the refractive index modulation (polymer/air) is 0.3. The average index is obtained by taking the average of the refractive indices of the polymer and air ($(1.6+1.0)/2=1.3$). FIG. 15A shows the diffraction efficiency versus input angle in air. FIG. 15B shows the diffractive efficiency versus wavelength.

FIGS. 16A-16B shows the corresponding diffraction efficiency characteristics of a grating formed from a polymer of index 1.8 (refractive index modulation 0.4) for the same beam angles in accordance with an embodiment of the invention. The higher index polymer results in an increase in the spectral bandwidth of the grating. The angular bandwidth (near 100%) covers all waveguiding angles. From consideration of the diffraction efficiency plots it should be apparent that the spectral bandwidth of the grating covers most of visible band. By considering the diffraction efficiency obtained for S and P polarized light the inventors have found that the DE characteristics do not vary significantly with polarization. Calculations also indicate that, in contrast to LC-extracted transmission gratings, neither the angular bandwidth nor the spectral bandwidth are affected by grating thickness.

Reflection Bragg gratings with K-vectors substantially normal to the waveguide substrates may present problems in the flushing out of LC since the extraction will need to take place through the edges of the grating. Such a grating will also be structurally unstable due the polymer regions not

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being supported. In many embodiments directed at providing waveguide-based displays, the reflection grating may be slanted allowing LC extraction to take place through the upper and lower faces of the grating. In some embodiments with K-vectors substantially normal to the waveguide substrates, the reflection Bragg grating can incorporate polymer scaffolding. FIGS. 17A-17E illustrate the step in fabricating such a reflection Bragg grating incorporating polymer scaffolding in accordance with an embodiment of the invention. In a first step conceptually illustrated in FIG. 17A, a grating structure 250A including alternating LC regions 252 and polymer regions 251 supported by a substrate 253 may be fabricated using a holographic exposure process as discussed above. Alternatively, a mask exposure process can be used. In a second step conceptually illustrated in FIG. 17B, the LC is extracted to provide the surface relief grating structure 250B in which the LC regions are now air-filled regions 254. In a third step conceptually illustrated in FIG. 17C, the grating can be refilled with a liquid crystal and monomer mixture 254a to produce a hybrid grating. In a fourth step conceptually illustrated in FIG. 17C, a multiplexed grating 250C combining a reflection grating (having K-vectors substantially normal to the substrate) and a transmission grating (having K-vectors substantially parallel to the plane of the substrate) is recorded into the mixture through an upper mask 255A and a lower mask 255B mask. The exposure illumination modulated by the masks is indicated by 256. Other arrangements of masks and illumination profiles can be used depending on the grating structures to be recorded. The exposed grating 250D conceptually illustrated in FIG. 17D includes horizontal LC regions 256A, horizontal polymer regions 256B, vertical LC regions 256C, and vertical polymer regions 256D which provide scaffolding for the horizontal polymer regions. In a final step conceptually illustrated in FIG. 17E, the LC is flushed out of the grating structure to form the finished grating 250E including the horizontal polymer grating elements 257A and vertical polymer grating elements 257B that have principal optical surfaces in contact with air and are supported by vertical polymer scaffolding elements 257C. In many embodiments, the reflection grating can have a thickness in the range 1-3 micron.

The process described in connection with FIGS. 17A-17E may produce various grating structures as illustrated in FIGS. 18-20.

FIG. 18 conceptually illustrates a reflection grating in accordance with an embodiment of the invention. The reflection grating 260 includes alternating first refractive index regions 261 and second refractive index regions 262.

FIG. 19 conceptually illustrates a reflection grating in accordance with an embodiment of the invention. The reflection grating 270 includes alternating first refractive index regions 271 and second refractive index regions 272. The reflection grating 270 further includes substantially vertical regions 273 of the second refractive index material.

FIG. 20 conceptually illustrates a combination reflection and transmission grating in accordance with an embodiment of the invention. The combination reflection and transmission grating 280 includes a reflection grating and a transmission grating. The combination grating 280 includes alternating first refractive index regions 281 and second refractive index regions 282. The combination grating 280 further includes approximately vertical regions 283 of a third refractive index material. The transmission grating is formed by the average of the first refractive index regions 281, second refractive index regions 282, vertical regions 283 of a third refractive index material. The diffraction of the

transmission grating is represented by the rays **284,285**. The reflection grating formed by the first refractive index regions **281** and second refractive index regions **282**. The diffraction of the reflection grating is represented by the rays **286,287**.

In many of the embodiments to be described below there is provided a waveguide-based display device including: a waveguide; a source of image modulated light projected over a field of view; an input coupler for coupling said light into a total internal reflection path within said waveguide; and a grating structure for providing beam expansion in at least one direction. The waveguide display can incorporate many of the embodiments discussed in the preceding paragraphs.

FIG. **21** conceptually illustrates a waveguide-based display in accordance with an embodiment of the invention. The waveguide-based display **290** includes a waveguide **291** supporting an input grating **292** and a grating structure **293**. Input light **294** having a wavelength less than or equal to a predefined threshold value is diffracted into a TIR path **295** while input light **296** having wavelength greater than the threshold value is not diffracted and emergence from the input grating without substantial deviation or attenuations as the zero-order light **297**. In some embodiments, the grating structure **293** can be designed to only diffract light below a threshold wavelength. The input grating **292** may include a short grating period that enables coupling of light below a threshold wavelength. In some embodiments, such as for example waveguide-based displays including layers operating on different primary colors, this feature can be used to limit color crosstalk between layers.

In many embodiments, a waveguide can support Bragg gratings including alternating regions formed from two different materials. FIG. **22** illustrates a plot **300** of different materials that have different dispersion (n - λ) curves **301, 302** which overlap at two or more points **303,304**. FIG. **23** conceptually illustrates a portion of a Bragg grating **310** formed from different dispersion materials **311,312** in accordance with an embodiment of the invention. The Bragg gratings may be used as the input grating **292** described in connection with FIG. **21** to provide waveguide selectivity. In some embodiments, the wavelength selectivity may be implemented in other gratings (e.g. fold gratings and output gratings) used in the waveguide to minimise the effects of cross talk between stacked RGB waveguides.

FIGS. **24-27** conceptually illustrate waveguide gratings with spatially varying diffraction efficiency in accordance with an embodiment of the invention. The gratings can be implemented as Bragg gratings, as SRGs, or as hybrid Bragg/SRG structures according to the principles discussed above. In many embodiments, the diffraction efficiency can be controlled by varying the grating depth and fill factor as a function of wavelength. FIG. **24** conceptually illustrates a grating in accordance with an embodiment of the invention. The grating **320** may include spatially varying modulation **321,322** at different positions.

FIG. **25** conceptually illustrates a SRG in accordance with an embodiment of the invention. The SRG **330** includes spatially varying modulation **321,322** as the grating **320** described in connection with FIG. **24**. The SRG **330** further includes at least partial backfilling with a backfilling material **331** of refractive index different to that of the previous grating **320**. As discussed above the SRG **330** can be formed by extracting liquid crystal from a HPDLC grating and then refilling with the backfill material **331**.

FIGS. **26-27** conceptually illustrates various plots disclosing the spatial variation of the refractive index modulation:

$\delta n(X)$, $\delta n(Y)$ **340,350**, along orthogonal directions (X,Y) for a grating designed to provide continuous spatial variation of index modulation **341,351**.

FIG. **28** conceptually illustrates a waveguide apparatus in accordance with an embodiment of the invention. The waveguide apparatus **360** including a waveguide **361** supporting an input grating **362** and a grating structure including a fold grating **363** configured such that ray paths **364,365** within the fold grating **363** have substantially the same geometrical path length in some embodiments. By designing the grating so that the path differences between the ray paths **364,365** are greater than the coherence length and the rays **364,365** can be summed incoherently to eliminate fringing and other illumination artefacts such as might occur in laser illuminated waveguides.

FIG. **29** conceptually illustrates a waveguide apparatus in accordance with an embodiment of the invention. The waveguide apparatus **370** includes a waveguide **371** supporting an input grating **372** having a spatially varying distribution of K-vectors **373** and a grating structure **374** having a spatially varying distribution of K-vectors **375** in some embodiments.

FIG. **30** conceptually illustrates a waveguide apparatus in accordance with an embodiment of the invention. The waveguide apparatus **380** comprising a waveguide **381** supporting an input grating **382**, a grating structure **383** (e.g. a fold grating), and at least one partially reflective coating **384**. In many embodiments the coating **384** can at least partially overlap the input grating **382**. In many embodiments, the coating **384** can at least partially overlap the grating structure **383**. In other embodiments, the coating **384** can at least partially overlap both of the input grating **382** and the grating structure **383**. One purpose of the coating **384** may be to overcome interference fringe formation in a laser illuminated waveguide.

FIG. **31** conceptually illustrates a portion of a grating structure in accordance with an embodiment of the invention. The grating structure **390A** includes non-diffracting regions **391A**, gratings of a first prescription **392A**, and gratings of a second prescription **393A** disposed in a non-repeating pattern.

FIG. **32** conceptually illustrates a portion of a grating structure in accordance with an embodiment of the invention. The grating structure **390B** includes a non-diffracting regions **391B**, gratings **392B** of a first prescription, and gratings **393B** of a second prescription disposed in a repeating pattern. The grating structures of FIGS. **31-32** can be used to provide two-dimensional beam expansion and uniform light extraction from a waveguide.

FIG. **33** conceptually illustrates a portion of a grating structure in accordance with an embodiment of the invention. The grating structure **390** includes alternating high index regions **391** and low index regions **392** with a grating pitch (p). As illustrated, the grating pitch p may be the distance between the peaks or centres of successive low index and high index regions. In many embodiments, the grating pitch p can be much greater than a wavelength λ to enable a wide field of view and low dispersion. In many embodiments, the grating pitch p can be in the range 5-1000 microns. The large pitch value results in many diffracted orders which can be used for optimising field of view and color dispersion, enabling single layer wide FOV full color waveguides. In many embodiments the high index regions and low index regions may have different widths.

FIG. **34** conceptually illustrates a waveguide apparatus **400** in accordance with an embodiment of the invention. The waveguide apparatus **400** includes a ray multiplying element

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404 disposed before an input coupler for debanding in accordance with an embodiment of the invention. In many embodiments, the input coupler 402 is a grating. In some embodiments, the ray multiplying element 404 can be disposed after the input coupler 402. The waveguide apparatus 400 includes a waveguide 401 supporting an input grating 402, a grating structure 403, and a ray multiplying element 404 for controlling beam width. An optical path through the waveguide in some embodiments is illustrated by the rays 405-407. In many embodiments, the ray multiplying element 404 can include mirrors, gratings and can have angle dependence and different beam width multiplying factors matched to different beam incidence angles and source exit pupils.

FIG. 35 conceptually illustrates a waveguide apparatus in accordance with an embodiment of the invention. The waveguide apparatus 410 uses a ray multiplying element 414 for debanding and an input prism coupler 412. The waveguide apparatus 410 includes a waveguide 411, a prism coupler 412, a grating structure 413, and a ray multiplying element 414. As illustrated, the ray multiplying element 414 may be positioned below the input prism coupler 412. An optical path through the waveguide 411 is illustrated by the rays 415-417.

FIG. 36A conceptually illustrates a plan view of a waveguide-based display in accordance with an embodiment of the invention. The display 420 uses a grating structure 423 including an array of grating elements 423A, 423B, 423C. The display 420 includes a waveguide 421 supporting an input coupler 422 and a grating structure 423 for beam expansion and extraction including an array of grating elements 423A, 423B. The array of grating elements include elements 423A for diffracting a first wavelength, elements 423B for diffracting a second wavelength and elements 423C for diffracting a third wavelength. In many embodiments, the gratings elements 423A, 423B, 423C of each wavelength can have different K-vectors. In many embodiments, the K-vectors can be selected from a group of two different K-vectors. In many embodiments, the K-vectors can be selected from a group of two opposing K-vectors. In some embodiments, in which the grating structure operates on red, green and blue light and requires two different K-vectors for performing two-dimensional beam expansion and extraction, a total of six different grating prescriptions are required. In several embodiments, the array may include non-diffracting regions. Input light can be directed into the waveguide 421 at different angle 81, 82. Examples of optical paths through the waveguide are illustrated by the rays 424A-426A and 424B-426B. The ray paths shown can, in some embodiments, represent light of different wavelengths. Light of a given input direction may undergo multiple grating interactions within the grating structure 423, undergoing diffraction only at gratings for which the incident ray lies within the spectral or angular bandwidth of the grating. Examples of optical paths through the grating structure is illustrated by the TIR ray 427, 428.

In some embodiments, the red, green and blue gratings can be exposed using a spatial light modulator array mask, illuminating the separate RGB array pattern on the patterned mask time sequentially to avoid crosstalk during exposure.

In many embodiments, angle and wavelength Bragg selectivity, spatial variations of modulation, thickness etc can be used to mitigate cross talk, control uniformity and minimise the effects of grating edge interactions.

In many embodiments the grating elements can be approximately 30 microns in size.

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FIG. 36B conceptually illustrates a plan view of a waveguide-based display in accordance with an embodiment of the invention. The waveguide-based display 430 uses a grating structure including an array of grating elements 433A, 433B, 433C in accordance with an embodiment of the invention. The waveguide-based display 430 includes a waveguide substrate 431 supporting three input coupler gratings 434A, 434B, 434C and a grating structure 433 for beam expansion and extraction including an array of grating elements. The array of grating elements includes elements 433A for diffracting a first wavelength, elements 433B for diffracting a second wavelength and elements 433C for diffracting a third wavelength. In many embodiments, the input couplers 434A, 434B, 434C diffract said first, second and third wavelengths respectively. An example of an optical paths through the grating structure is illustrated by the TIR ray 437. For example, the optical path of light coupled into the waveguide by the coupler 434A is illustrated by the rays 434-436.

FIG. 36C conceptually illustrates a plan view of a waveguide-based display in accordance with an embodiment of the invention. The waveguide-based display 440 includes a grating structure 443 including an array of grating elements 443A, 443B, 443C. The waveguide-based display 440 includes a waveguide substrate 441 supporting an input coupler grating array 442 including grating elements 444A, 444B, 444C for coupling first, second and third wavelength light into TIR paths within the waveguide, and a grating structure 443 for beam expansion and extraction including an array of grating elements including, in many embodiments, elements 443A for diffracting the first wavelength, elements 443B for diffracting the second wavelength and elements 443C for diffracting the third wavelength. An example of an optical paths through the grating structure is illustrated by the TIR ray 447. The optical path of light coupled into the waveguide is illustrated by the rays 445-446.

FIG. 37 conceptually illustrates a side elevation view of a waveguide-based display 450 based on the embodiments of FIGS. 36A, 36B, 36C. The waveguide-based display 450 includes a waveguide 451 with an input coupler provided by a prism 454 in accordance with an embodiment of the invention. The waveguide-based display 450 further includes a grating structure 453. The prism 454 couples incident light 455 into TIR paths within the waveguide. The grating structure 453 performs beam expansion and extracts the light 456. FIG. 38 illustrates a plan view of the waveguide-based display described in connection with FIG. 37.

FIG. 39 conceptually illustrates an apparatus 470 for recording a grating structure including a grating array in accordance with an embodiment of the invention. The apparatus 470 includes a substrate 471 supporting a layer of holographic recording material, a master grating 472, red, green and blue lasers 473A, 473B, 473C emitting beams 474A, 474B, 474C, which can be combined by a stack of dichroic filters 475A, 475B, 475C into the common beam direction into the collimating/beam expansion optics 476. The expanded beam may be diffracted by the master grating 472 to form a diffracted beam 477 and a zero-order beam 478 used for copying the master grating 472 into the holographic material layer. Many other configurations for combining, collimating and beam expanding the laser beams should be apparent to those skilled in the art. For example, holographic recording material with red, green, or blue sensitized dyes can be deposited onto areas designated the red, green and blue gratings respectively. In many embodiments, the holographic recording material is a HPDLC

material system. In many embodiments, the grating regions are coated using an inkjet printer. In many embodiments, the grating regions can be coated with a spatial resolution of 30 microns. In many embodiments the gratings have rolled K-vectors. In many embodiments, the gratings are recorded using rapid pulse laser exposure so that complete red green and blue gratings exposure occurs within substantially the same time window. In many embodiments, the master grating **472** supports a hard coat to allow it to be positioned close to the recording material layer. In many embodiments, a 10 nm hard coat can be used for this purpose. In many embodiments, the master grating **472** is coated with a 50-micron cover glass or polymer overcoating. In many embodiments, the principles of the embodiment of FIG. **39** can be applied in a two-color band (blue-green and red, for example) waveguide.

In many embodiments, the grating structure can include a plurality of grating elements tiled in a repeating pattern. In many embodiments, the grating structure can include a plurality of grating elements tiled in a non-repeating pattern. In many embodiments, the grating structure can include grating elements of non-rectangular shape. In many embodiments, the grating structure can have a spatially varying refractive index modulation. In many embodiments, the grating structure can include a plurality of grating elements with spectral response selected from a group containing at least two different spectral responses. In many embodiments, the grating structure can include a plurality of grating elements with angular response selected from a group containing at least two different angular responses. In many embodiments, the grating structure can include a plurality of grating elements with birefringence including at least two different birefringence characteristics. In many embodiments, the grating structure can include a plurality of grating elements with polarization response including at least two different polarization responses. In many embodiments, the grating structure can include a plurality of grating elements with refractive index modulation including at least two different refractive index modulations. In many embodiments, the grating structure can include a plurality of grating elements with average refractive index including at least two different refractive indices. In many embodiments, the grating structure can include a plurality of grating elements with grating thickness including at least two different grating thicknesses. In many embodiments, the grating structure can include a plurality of grating elements having grating prescriptions including at least two different grating prescriptions. In many embodiments, the grating structure can include a plurality of grating elements having K-vectors including at least two different K-vectors. In many embodiments, the grating structure can include a plurality of grating elements including diffracting grating elements arrayed with non-diffracting grating elements. In many embodiments, the grating structure can include a plurality of grating elements recorded in material including at least two different materials. In many embodiments, the grating structure can include a plurality of grating elements recorded in material having dispersion curves including at least two different dispersion curves. In many embodiments, the grating structure can include a plurality of grating elements recorded in material having wavelength sensitivity selected from a group including at least two different wavelength sensitivities. In many embodiments, the grating structure can include a plurality of grating elements recorded in material having holographic exposure time including at least two different holographic exposure times. In many embodiments, the grating structure can include a plurality of grating elements

recorded in material having holographic exposure energy including at least two different holographic exposure energies. In many embodiments, the grating structure is configured to support ray paths differing by a distance shorter than the coherence length of the source. In many embodiments, the grating structure can include at least one grating having a grating pitch much greater than a wavelength of light. In many embodiments, the grating structure can include at least one grating having a spatially variation of: spatial variation of grating thickness; average refractive index; refractive index modulation, or birefringence. In many embodiments, the grating structure can include rolled K-vector grating, a dual interaction grating, a multiplexed grating, a lossy grating, a sub wavelength grating, and/or a chirped grating. In many embodiments, the grating structure can include at least one surface relief grating formed by liquid crystal extraction from a grating recorded in HPDLC. In many embodiments, the grating structure can include at least one grating formed by at least partially backfilling a surface relief grating formed by liquid crystal extraction from a HPDLC recorded grating with another material having a refractive index higher than that of the extracted liquid crystal. In many embodiments, the grating structure can include at least one grating formed by at least partially backfilling a surface relief grating formed by liquid crystal extraction from a HPDLC recorded grating with another material having a refractive index lower than that of the extracted liquid crystal. In many embodiments, the grating structure can include at least one reflection grating. In many embodiments, the grating structure can include at least one transmission grating. In many embodiments, the grating structure can be a switching grating. In many embodiments, the grating structure can further include the group of a beamsplitter layer, an anti-reflection coating, an optical bandpass filter, a polarization modification layer, and/or an alignment layer. In many embodiments, the grating structure can provide beam expansion and extraction of red, green and blue image modulated light over a field of view. In many other embodiments, the image modulated light is monochromatic. In many embodiments, the grating structure can be recorded into a material including a mixture including a liquid crystal and at least one monomer. The mixture may include at least one blue phase liquid crystal and at least one monomer. In many embodiments, the input coupler can be a grating or a prism. In many embodiments, the input coupler can include a plurality of gratings. In many embodiments, the input coupler can be a switching grating.

In many embodiments, a photonic crystal formed by liquid crystal extraction may be used to form a multiplexed grating. FIG. **40** illustrates a multiplexed grating structure in accordance with an embodiment of the invention. The multiplexed grating may be manufactured using a method. In a first step, a first mixture of liquid crystal is provided. A first grating **482** including alternating liquid crystal and polymer regions can be formed in the first mixture using a holographic exposure. The LC regions can be flushed to form a first set of polymer regions separated by air. A second mixture of liquid crystal can be provided in the air regions of the first grating. A second grating **484** including alternating liquid crystal and polymer regions can be formed in the second mixture using a holographic exposure. The LC regions can be flushed to form polymer regions separated by air. A third mixture of liquid crystal can be provided in the air regions formed by the first and second gratings. A third grating **486** comprising alternating liquid crystal and polymer regions can be formed in the second mixture using a holographic exposure. The LC regions can be flushed to

form polymer regions separated by air. FIG. 40 shows the final grating structure formed by the above process. The first grating 482, the second grating 484, and the third grating 486 represent the polymer regions of the final grating. In many embodiments, the three superimposed gratings 482, 484, 486 may have the same refractive index. Air spaces 488 remaining after the above process has been completed. The structure of FIG. 40 multiplexes three gratings 482, 484, 486. In some embodiments, the three gratings 482, 484, 486 may be formed from different monomers to provide a required spatial refractive index modulation variation. In some embodiments, the air regions 488 can be backfilled with an optical material for providing a desired refractive index contrast. It should be apparent from consideration of the embodiment of FIG. 40 that a similar process can be applied to multiplexing any number of grating structures subject to material and process limitations.

One important advantage of the LC evacuated gratings is that they will not clear at elevated temperature so have advantage for automotive use, or any other higher temperature environment use. The LC evacuated grating principle can be applied to gratings of any scale. In some embodiments, the LC evacuated grating has spatially varying diffraction efficiency. The application of multiplexing, spatial varying thickness, k-vector directions, and diffraction efficiency in the present invention is based on the embodiments, drawings and teachings provided.

In one embodiment the gratings are recorded in uniform modulation liquid crystal-polymer material system such as the ones disclosed in United State Patent Application Publication No.: US2007/0019152 by Caputo et al. and PCT Application No.: PCT/EP2005/006950 by Stumpe et al. both of which are incorporated herein by reference in their entireties. Uniform modulation gratings are characterized by high refractive index modulation (and hence high diffraction efficiency) and low scatter.

The invention can be applied using grating arrays made up of grating elements of many different geometries that are limited only by geometrical constraints and the practical issues in implementing the arrays. In many embodiments, the grating array can include grating elements that are aperiodic (non-repeating). In such embodiments, the asymmetry in the geometry and the distribution of the grating elements can be used to produce uniformity in the output illumination from the waveguide. The optimal grating elements sizes and geometries can be determined using reverse vector raytracing from the eyepiece through the output and input gratings (and fold gratings, if used) onto the grating array. A variety of asymmetric grating element patterns can be used in the invention. For example, FIG. 41A conceptually illustrates a portion 490 of a grating element pattern including rectangular elements 490A-490F of differing size and aspect ratio for use in an emissive display panel in accordance with an embodiment of the invention. In some embodiments, the grating element array can be based on a non-repeating pattern based on a finite set of polygonal base elements. FIG. 41B conceptually illustrates a portion 500 of a grating element pattern including Penrose tiles 500A-500J in accordance with an embodiment of the invention. The tiles can be based on the principles disclosed in U.S. Pat. No. 4,133,152 by Penrose entitled "Set of tiles for covering a surface" which is hereby incorporated by reference in its entirety. Patterns occurring in nature, of which honeycombs are well known examples, can also be used in many embodiments.

In many embodiments, the grating element can be configured in arrays of identical regular polygons. For example,

FIG. 41C conceptually illustrates a portion 510 of a grating element pattern including hexagons 510A-510C in accordance with an embodiment of the invention. FIG. 41D conceptually illustrates a portion 520 of a grating element pattern including squares 520A-520C in accordance with an embodiment of the invention. FIG. 41E conceptually illustrates a portion 530 of a grating element pattern comprising diamond-shaped elements 530A-530D in accordance with an embodiment of the invention. FIG. 41F conceptually illustrates a portion 540 of a grating element pattern including isosceles triangles 540A-540H in accordance with an embodiment of the invention.

In many embodiments, the grating elements can have vertically or horizontally biased aspect ratios. FIG. 41G conceptually illustrates a portion 550 of a grating element pattern including hexagons 550A-550C of horizontally biased aspect ratio in one embodiment. FIG. 41H conceptually illustrates a portion 560 of a grating element pattern including rectangles 560A-560D of horizontally biased aspect ratio in accordance with an embodiment of the invention. FIG. 41I conceptually illustrates a portion 570 of a grating element pattern including diamond shaped elements 570A-570D of horizontally biased aspect ratio in accordance with an embodiment of the invention. FIG. 41J conceptually illustrates a portion 580 of a grating element pattern including triangles 580A-580H of horizontally biased aspect ratio in accordance with an embodiment of the invention.

In many embodiments, using the grating element configurations discussed above, the grating elements can have differing spectral response. In many embodiments, using the grating element configurations discussed above, the grating elements can have differing angular response. In many embodiments, using the grating element configurations discussed above, the grating elements can have differing polarization response. In many embodiments, the grating element can have spectral, angular, and/or polarization response varying spatially across the grating element array. In many embodiments, grating elements of different sizes and geometries and different spectral, angular and polarization response can be arranged to provide a spatial emission variation for controlling uniformity in the final image.

FIG. 42 conceptually illustrates a method for forming a waveguide color grating structure in accordance with an embodiment of the invention. As shown, the method 590 of forming a grating structure is provided. Referring to the flow diagram, the method 590 includes providing (591) a waveguide substrate. A master grating patterned into an array comprising red, green and blue diffracting grating elements can be provided (592). Coats of red, green and blue sensitized holographic recording material can be provided in the designated red, green and blue grating regions of the substrate correspond to the master grating array can be deposited (593). Red, green and blue laser sources can be provided (594). Red, green and blue exposure can be applied (595) to the holographic recording material using diffracted and zero order beams from the master. A grating structure comprising an array of red, green and blue differing gratings can be recorded (596).

In many embodiments, the holographic recording material can be a mixture of at least one liquid crystal and at least one monomer. In many embodiments, the holographic recording material can further include at least one component selected from the group of a photoinitiator, a dye, a nano particle, a surfactant, a blue phase liquid crystal and a reactive mesogen.

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In many embodiments, the master is an amplitude grating with a hard overcoat. In many embodiments, the coated substrate is protected by a thin cover glass or a polymer layer. In many embodiments, the distance from the amplitude grating to the holographic recording plane is in the range 1-50 micron to avoid trichromatic pixel exposure overlap. In many embodiments, the grating elements thickness is approximately 2 microns. In many embodiments, the grating elements are printed with a resolution and accuracy of approximately 30 microns.

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While the above description contains many specific embodiments of the invention, these should not be construed as limitations on the scope of the invention, but rather as an example of one embodiment thereof. It is therefore to be understood that the present invention may be practiced in ways other than specifically described, without departing from the scope and spirit of the present invention. Thus, embodiments of the present invention should be considered in all respects as illustrative and not restrictive. Accordingly, the scope of the invention should be determined not by the embodiments illustrated, but by the appended claims and their equivalents.

What is claimed is:

1. A waveguide-based display device comprising:
a waveguide;
a source of image modulated light projected over a field of view;
an input coupler for coupling said light into a total internal reflection (TIR) path within the waveguide; and
a grating structure for providing beam expansion of the TIR light in at least one direction and extracting the TIR light from the waveguide, wherein the grating structure comprises a plurality of grating elements having at least two different grating prescriptions and at least one surface relief grating formed by a phase separation process;
wherein the grating structure is configured to enable a plurality of ray path lengths within the grating structure that differ in length by a distance shorter than a coherence length of the source of image modulated light.
2. The waveguide-based display device of claim 1, wherein the plurality of grating elements is tiled in a repeating pattern or non-repeating pattern.
3. The waveguide-based display device of claim 1, wherein the grating structure further comprises a spatial refractive index modulation.
4. The waveguide-based display device of claim 1, wherein the plurality of grating elements comprises at least two different spectral responses, at least two different angular responses, at least two different birefringence characteristics, at least two different polarization responses, at least two different refractive index modulations, an average refractive index including at least two different refractive indices, at least two different grating thicknesses, and at least two different K-vectors.
5. The waveguide-based display device of claim 1, wherein the plurality of grating elements comprises diffracting grating elements arrayed with non-diffracting grating elements.

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6. The waveguide-based display device of claim 1, wherein the plurality of grating elements is recorded in material containing at least two different materials.

7. The waveguide-based display device of claim 1, wherein the plurality of grating elements are recorded in material having dispersion curves selected from a group containing at least two different dispersion curves.

8. The waveguide-based display device of claim 1, wherein the plurality of grating elements is recorded in material having at least two different wavelength sensitivities, at least two different holographic exposure times, or at least two different holographic exposure energies.

9. The waveguide-based display device of claim 1, wherein the plurality of grating elements comprises at least one grating pitch greater than a wavelength of the TIR light.

10. The waveguide-based display device of claim 1, wherein the plurality of grating elements comprises a spatial variation of at least one selected from the group consisting of: spatial variation of grating thickness, average refractive index, refractive index modulation, and birefringence.

11. The waveguide-based display device of claim 1, wherein the grating structure comprises at least one selected from a group consisting of: a rolled K-vector grating, a dual interaction grating, a multiplexed grating, a lossy grating, a sub wavelength grating, and a chirped grating.

12. The waveguide-based display device of claim 1, wherein the at least one surface relief grating is formed by liquid crystal extraction from a grating recorded in holographic polymer dispersed liquid crystal (HPDLC).

13. The waveguide-based display device of claim 1, wherein the at least one surface relief grating is formed by at least partially backfilling a surface relief grating formed by liquid crystal extraction from a holographic polymer dispersed liquid crystal (HPDLC) recorded grating with another material having a refractive index higher than or lower than that of the extracted liquid crystal.

14. The waveguide-based display device of claim 1, wherein the grating structure comprises at least one reflection grating or at least one transmission grating.

15. The waveguide-based display device of claim 1, wherein the grating structure provides beam expansion and extraction of red, green and blue image modulated light over a field of view.

16. The waveguide-based display device of claim 1, wherein the grating structure is recorded into a material selected from the group of a mixture comprising at least one liquid crystal and at least one monomer, a mixture comprising at least one blue phase liquid crystal and at least one monomer.

17. The waveguide-based display device of claim 1, wherein the input coupler is a grating or a prism.

18. The waveguide-based display device of claim 1, wherein the at least one surface relief grating comprises a grating modulation depth greater than a grating pitch.

19. The waveguide-based display device of claim 1, further comprising an alignment layer.

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